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TSH suppression after thyroidectomy for thyroid cancer and bone health in postmenopausal women: A systematic review and meta-analysis

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TSH suppression after thyroidectomy for thyroid cancer and bone health in postmenopausal women: A systematic review and meta-analysis

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Abstract

Objective: The association between different levels of thyroid-stimulating hormone (TSH) suppression after thyroidectomy for thyroid cancer and bone health in postmenopausal women has not been fully investigated. We assessed the effects of TSH suppression on bone mineral density (BMD) in postmenopausal women undergoing thyroidectomy for thyroid cancer.

Research design and methods: PubMed, EMBASE, and Cochrane Library databases were searched (inception to April 10, 2020), and studies evaluating BMD in postmenopausal women with thyroid cancer undergoing thyroidectomy and levothyroxine administration were included. Differences in BMD were presented as standardized mean differences (SMDs), and meta-analyses were conducted using a random-effects model.

Results: In the analysis of 16 case-control studies including 426 patients and 701 controls without thyroid cancer, stringent TSH suppression (TSH, <0.10 mIU/L) after thyroidectomy showed deleterious effects on the BMD of the lumbar spine in postmenopausal women compared to controls (SMD, -0.55; 95% confidence interval [CI], -0.99 to -0.10; $I^2 = 75.8\%$). However, there was no significant difference in patients with moderate TSH suppression (TSH, 0.10–0.49 mIU/L). TSH suppression in postmenopausal women was not significantly associated with lower BMD of the femoral neck. In the subgroup analysis using lumbar spine data, total thyroidectomy (SMD, -0.89; 95% CI, -1.48 to -0.31) and baseline body mass index (BMI) <25 kg/m^2 (SMD, -0.67; 95% CI, -1.16 to -0.19) were linked to a significant decrease in postoperative BMD compared with near-total thyroidectomy and baseline BMI ≥25 kg/m^2.

Conclusions: Our meta-analysis showed that stringent TSH suppression had deleterious effects on the BMD of the lumbar spine after thyroidectomy for thyroid cancer in postmenopausal women. Further studies are needed to determine whether stringent TSH suppression after thyroidectomy for thyroid cancer increases the risk of fractures.

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PROSPERO registration: CRD42018104282

Keywords: thyroid cancer, thyroidectomy, bone mineral density, thyroid-stimulating hormone, thyroid-stimulating hormone suppression

Strengths and limitations of this study

- We performed a thorough literature search and strict quality assessment of the included studies to strengthen this systematic review and meta-analysis.
- This study is the first to qualitatively collate and assess the results of studies investigating the effect of TSH suppression intensity after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal women.
- A limitation of this study is that only a few studies with prospective design were included, which decreased the level of evidence; moreover, the effects of notable confounding factors and possible fluctuations in the postoperative TSH levels were not reflected in the analysis.

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INTRODUCTION

Thyroid stimulating hormone (TSH) levels are suppressed after thyroidectomy for thyroid cancer.[1-3] It is necessary to assess the risk–benefit and adverse effect of TSH suppression because patients with thyroid cancer have a favorable prognosis.[4] To prevent cancer recurrence and reduce cancer-related mortality, levothyroxine is used to achieve TSH suppression[5] as it inhibits the production of circulating TSH, a factor known to stimulate the proliferation of thyrocytes.[6] However, supraphysiological doses of exogenous levothyroxine may cause side effects, especially those related to postoperative bone health.[2,7] Endogenous hyperthyroidism increases the risk of osteoporosis and osteoporotic fractures[8,9]; however, the effects of TSH suppression due to levothyroxine administration after thyroidectomy are unclear.[10]

Although different levels and durations of TSH suppression are recommended post-thyroidectomydepending on the risk of thyroid cancer recurrence [1,11,12], the impact of varying levels of TSH suppression on bone health is unclear. Recent studies have evaluated the association between TSH suppression level and fractures.[8,13] However, it is difficult to determine the risk of bone loss induced by various levels of TSH suppression after thyroidectomy because a paucity of literature. Several literature reviews have reported that suppressed TSH levels led to osteoporosis and decreased bone mineral density (BMD) after thyroidectomy for thyroid cancer in postmenopausal women.[14-18] However, no studies have investigated the effects of varying TSH levels on bone health.

In this study, we performed a systematic review and meta-analysis of existing studies to investigate the effects of different levels of TSH suppression after thyroidectomy on postoperative BMD in postmenopausal women with thyroid cancer.

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MATERIALS AND METHODS

An expert panel was formed to define the scope of the meta-analysis, literature search, eligibility criteria, and statistical methods according to the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines.[19] The protocol of this systematic review and meta-analysis was registered in the PROSPERO database (registration number, CRD42018104282).

Literature search

A trained librarian searched the PubMed, EMBASE, and Cochrane Library databases to identify relevant studies published before April 10, 2020 (Supplemental Table 1). The following search terms, adapted to each database, were used: "*thyroidectomy*," "*levothyroxine*," "*thyroid stimulating hormone*," "*bone density*," "*osteoporosis*," and "*bone fractures*." We also included studied reported in earlier systematic reviews[14-17] and those identified from the bibliography of the included articles.

Eligibility criteria and study selection

The inclusion criteria were as follows: (1) all participants underwent thyroidectomy for thyroid cancer, followed by the administration of levothyroxine; (2) availability of BMD data, irrespective of whether this was the primary endpoint; (3) availability of data on the lumbar spine and femoral neck; (4) comparison of postoperative outcomes between the study and control groups; and (5) the most recent or informative publication data, in the case of multiple publications from the same population. The exclusion criteria were as follows: (1) only abstract available; (2) case reports, reviews, and editorials; (3) studies with pediatric or pregnant populations; (4) publication language other than English; and (5) missing measures of precision (standard deviations or standard errors) when effect estimates were presented.

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Two authors (D.K. and Y.K.) independently assessed the eligibility of all studies identified through the electronic literature search based on the titles and abstracts. Eligible articles with full texts were further evaluated for inclusion. Disagreement was resolved by consensus.

Data extraction and risk-of-bias assessment

A standard template suggested by Cochrane was adapted and applied for data extraction.[20] Two authors (D.K. and J.H.) independently extracted data, including the first author's name; year of publication; study location; study design; age, body mass index (BMI), race, and number of the study participants; etiology; extent of thyroidectomy; location and value of the BMD measurement; follow-up duration after thyroidectomy; levothyroxine dose; preoperative and postoperative laboratory results including TSH, free thyroxine, parathyroid hormone, and calcium levels; medications used by the study participants; and medical condition of the study participants.

We adapted existing quality assessment tools[21-23] to evaluate the quality of the included studies. We selected three items related to the risk of bias (study design, sampling method, and confounders control) and one item each for generalizability (multisite vs. single site) and statistical power (sample size).

Data synthesis and analysis

Each outcome was presented with estimates calculated as standardized mean differences (SMDs). Assuming that the baseline values of patients and controls without thyroid cancer were balanced and the confounders were controlled by the study design, we compared the measurements at the end of the follow-up period. SMDs were coded so that positive values represented a higher BMD in the thyroidectomy group than in the control

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group, with 0.2, 0.5, and 0.8 corresponding to small, moderate, and large effects, respectively.[24] We estimated the pooled SMDs and 95% confidence intervals (CIs) using the DerSimonian and Laird model. A two-tailed p-value of <0.05 was considered statistically significant. We assessed heterogeneity using the Mantel–Haenszel model and I^2 values (the percentage of variance in the pooled estimate due to between-study differences), with I^2 thresholds of ≤25% (low), 26%–50% (moderate), and >50% (high).

Subgroup analyses were performed considering the TSH level (categorized as stringent suppression, <0.10 mIU/L; moderate suppression, 0.10–0.49 mIU/L; and lower normal, 0.50–2.0 mIU/L). Other investigated factors included the follow-up duration, baseline age, baseline BMI, race, and extent of thyroidectomy (total vs. near-total). We assessed publication bias based on funnel plot asymmetry using the Begg's and Egger's tests at $p < 0.1$. Stata version 13.1 software (StataCorp., College Station, TX, USA) was used for all statistical analyses.

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RESULTS

Studies included in the meta-analysis

The systematic literature search retrieved a total of 1,470 studies (Supplemental Figure 1). We further included two studied from the bibliography of the included articles. Of the included studies, 505 duplicate studies were excluded. The remaining 967 articles were screened based on the title and abstract and assessed for eligibility. The full text of 157 potentially eligible studies was evaluated by two independent authors (D.K. and J.H.), of which 140 studies did not meet the inclusion criteria (reasons for exclusion are summarized in Supplemental Figure 1). A total of 16 studies[25-41] including 1,127 postmenopausal women met the inclusion criteria.

Table 1 lists the summary and characteristics of each study.

TABLE 1. Characteristics of studies included in the analysis

First author (year, location)	Study population				Methods		Outcomes
	Age, mean (SD), years	Etiology	Extent of thyroidectomy	No. of postmenopausal participants, n (total=1,417)	Assessment tool of bone density (measurement site included in the meta-analysis)	Outcomes included in the meta-analysis (unit)	Assessment time after thyroidectomy, mean, year
Recker (1989, USA)	50.2	DTC	Total	Patients, 18; Controls, 16	DPA (lumbar spine and femoral regions)	BMD (g/m^2)	9.6
Diamond (1991, Australia)	59 (2.8)	DTC	Total	Patients, 10; Controls, 10	DPA (Lumbar spine, femoral neck)	BMD (g/m^2)	5.9
Franklyn (1992, UK)	63.4 (8.1)	DTC	Subtotal	Patients, 26; Controls, 26	DEXA (lumbar spine and femoral regions)	BMD (g/m^2)	NR
Abugassa (1993, Sweden)	58 (2)	Thyroid cancer	Total	Patients, 13; Controls, 12	DPA (lumbar spine and femoral regions)	BMD (g/cm^2)	13.0

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Kung (1993, China)	62 (8)	Thyroid cancer	Total	Patient, 34; control, 34	DEXA (lumbar spine, femoral regions)	BMD (g/cm^2)	12.2
Giannini (1994, Italy)	Patients, 57.6 (1.7); Control, 56 (0.4)	Non-medullary thyroid carcinoma	Total	Patients, 13; Controls, 11	DEXA (lumbar spine, femoral regions)	BMD (g/cm^2)	7.6
Hawkins (1994, Spain)	Patients, 59.6 (7.5); Control, 49.3 (5.2)	Thyroid cancer	Near total	Patients, 21; Controls, 53	DEXA (lumbar spine)	BMD (g/cm^2)	6.2
Frusciante (1998, Italy)	Patients, 55.3 (1.7); Controls, 53.3 (2.5)	DTC	Near total	Patients, 6; Controls, 7	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.7
Toivonen (1998, Finland)	Patients, 45 (range, 27 to 71); Controls, 43 (range, 26 to 65) [a]	DTC	Near total	Patients, 10; Controls, 12	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	10
Goerres (1998, Switzerland)	Patients, 61.6 (12.0); Controls, 562.7 (8.1)	Thyroid cancer	Total	Patients, 23; Controls, 42	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	8.8
Chen (2004, China)	57.7 (6.9)	DTC	Near total	Patients, 25; Controls, 25	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	7.8
Reverter (2005, Spain)	58 (9)	DTC	Near total	Patients, 44; Controls, 44	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	12.0
Eftekhari (2008, Iran)	Patients, 51.7 (7.3); Controls, 51.2 (7.8) [a]	DTC	NR	Patients, 33; Controls, 33	DEXA (lumbar spine)	BMD (g/cm^2)	1.2

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Tournis (2015, Greece)	Patients, 56.7 (5.9); Controls, 57.8 (6.1)	DTC	Total	Patients, 40; Controls, 60	DEXA (lumbar spine, femoral neck)	BMD (mg/cm^3) [b]	5.0
Moon (2016, Korea)	Patients, 61.4 (7.6); Controls, 61.4 (7.6)	DTC	NR	Patients, 74; Controls, 222	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.2
Kim (2019, Korea)	Patients, 53; Controls, 55	DTC	Total	Patients, 36; Controls, 94	DEXA (lumbar spine)	BMD (g/cm^2)	4.9

SD = standard deviation; NR = not reported; DPA = dual photon absorptiometry; DEXA = dual-energy x-ray absorptiometry; DTC = differentiated thyroid carcinoma; BMD = bone mineral density
[a] These studies presented the mean or median age of total study population including both pre- and postmenopausal women.
[b] This value was calculated with the method of volumetric BMD. For more information, refer to Tournis et al. (2015).

Studies were published as early as 1989, and the follow-up duration after thyroidectomy ranged from 1 to 15 years. Three studies[26,37,40] were prospective studies, and three studies[39,38,30] involved consecutive sampling (Table 2).

TABLE 2. Quality rating for studies included in the analysis

First author (Year)	**Design**	**Sampling strategy**	**Multisite**	**Sample size**	**Method of controlling for confounders**	**Confounders included in the analysis** (Confounders included in the exclusion criteria are presented in the right two columns)	**Exclusion criteria for medication** [a]	**Exclusion criteria for medical conditions** [b]
McDermott (1983)	Retrospective cohort with matched control	Nonconsecutive	No	36	Design (Matching, Exclusion)	Matching: Age, sex, height, weight	Stated with the list of drugs	Stated with the list of medical conditions
Recker (1989)	Retrospective cohort with	Nonconsecutive	No	34	Design (Matching, Exclusion)	NR	Stated with the list of drugs	Stated with the list of medical

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	matched control							conditions
Diamond (1991)	Prospective cohort with matched control	Nonconsecutive	No	48	Design (Matching, Exclusion)	Matching: Age, sex, menopausal status	Stated with the list of drugs	Stated
Schneider (1991)	Retrospective cohort with matched control	Nonconsecutive	No	63	Design (Matching, Exclusion)	Matching: Age, sex, menopausal status	Stated	Stated
Stepan (1992)	Retrospective cohort with matched control	Nonconsecutive	No	116	Design (Matching, Exclusion)	Matching: Age, sex, weight, height, menopausal status	Stated	Stated
Franklyn (1992)	Retrospective cohort with matched control	Nonconsecutive	No	98	Design (Matching, Exclusion)	Matching: Age, sex, BMI, menopausal status, smoking history, calcium intake score	Stated with the list of drugs	Stated with the list of medical conditions
Pioli (1992)	Retrospective cohort with matched control	Consecutive	No	38	Design (Matching, Exclusion)	Matching: Age, sex, menopausal status	Stated	Stated
Abugassa (1993)	Retrospective cohort with matched control	Consecutive	No	25	Design (Matching, Exclusion)	Matching: Age, sex, menopausal status, postoperative time, postmenopausal age	Stated with the list of drugs	Stated with the list of medical conditions
Kung (1993)	Retrospective cohort with matched control	Nonconsecutive	No	68	Design (Matching, Exclusion)	NR	Stated with the list of drugs	Stated with the list of medical conditions
Giannini (1994)	Retrospective cohort with matched control	Nonconsecutive	No	46	Design (Matching, Exclusion)	Matching: Sex, height, weight, menopausal status	Stated	Stated
Hawkins (1994)	Retrospective cohort	Nonconsecutive	No	74	Design (Matching,	Matching: Age, sex,	Stated with the	Stated

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	with matched control				Exclusion)	menopausal status	list of drugs	
Fruscian te (1998)	Retrospec tive cohort with matched control	Nonconsec utive	No	39	Design (Matchin g, Exclusion)	Matching: Age, Sex, BMI, menopausal status	Stated	Stated
Toivone n (1998)	Retrospec tive cohort with matched control	Nonconsec utive	No	67	Design (Matchin g, Exclusion)	Matching: Age, Sex, menopausal status	Stated	Stated
Goerres (1998)	Retrospec tive cohort with matched control	Consecutiv e	No	114	Design (Matchin g, Exclusion)	Matching: Age, Sex	Stated with the list of drugs	NR
Chen (2004)	Retrospec tive cohort with matched control	Nonconsec utive	No	50	Design (Matchin g, Exclusion)	Matching: Age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditio ns
Reverter (2005)	Prospecti ve cohort with matched control	Nonconsec utive	No	88	Design (Matchin g, Exclusion)	Matching: Age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditio ns
Sajjinan ont (2005)	Retrospec tive cohort with matched control	Nonconsec utive	No	44	Design (Matchin g, Exclusion)	Matching: Sex, menopausal status	Stated with the list of drugs	Stated with the list of medical conditio ns
Eftekhar i (2008)	Retrospec tive cohort with matched control	Consecutiv e	No	132	Design (Matchin g, Exclusion)	Matching: Age, sex, menopausal status	Stated with the list of drugs	Stated with the list of medical conditio ns
Baqi (2010)	Retrospec tive cohort	Nonconsec utive	No	70	Design (Matchin g, Exclusion) [c]	Matching: Age, sex, menopausal status	Stated with the list of drugs	Stated with the list of medical conditio ns
Schneid er (2012)	Prospecti ve cohort	Consecutiv e	No	160	Design (Exclusio n)	NR	Stated with the list of drugs	Stated with the list of medical conditio ns

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Tournis (2015)	Prospective cohort with matched control	Nonconsecutive	No	169	Design (Matching, Exclusion)	Matching: Age, sex, BMI, menopausal status, calcium intake, years since menopause in case of postmenopausal women	Stated with the list of drugs	Stated with the list of medical conditions
Moon (2016)	Retrospective cohort with matched control	Nonconsecutive	No	496	Design (Matching, Exclusion)	Matching: Age, sex	Stated with the list of drugs	Stated with the list of medical conditions

NR = not reported; NA = not applicable; BMI = body mass index; TSH = thyroid stimulating hormone; BMD = bone mineral density

[a] Categories include NR (not reported), Stated (without the list of drugs), and Stated with the list of drugs. The lists of drugs for exclusion are summarized in Supplemental Table 2.

[b] Categories include NR (not reported), Stated (without the list of drugs), and Stated with the list of drugs. The lists of medical conditions for exclusion are summarized in Supplemental Table 2.

[c] Additional analyses used multivariable regression to evaluate the association between TSH level and bone density after thyroidectomy controlled for TSH, free thyroxine, urinary calcium, serum alkaline phosphatase, serum osteocalcin, serum telopeptide, BMD of the lumbar spine, and BMD of the femoral hip. The comparison of bone density between patients and controls was performed using a paired *t*-test without controlling for confounders.

All studies, except one,[39] controlled for confounders in the study design using matching factors and the exclusion criteria (Table 2 and Supplemental Table 2). The mean TSH levels after thyroidectomy ranged between 0.00 and 1.8 mIU/L; the postoperative levothyroxine doses are summarized in Supplemental Table 2.

Effects of TSH suppression on the lumbar spine

All the included studies reported BMD of the lumbar spine in 426 postmenopausal women undergoing thyroidectomy for thyroid cancer and 701 controls. Overall, TSH suppression after thyroidectomy showed detrimental effects on the BMD of the lumbar spine (SMD, -0.46; 95% CI, -0.77 to -0.16; $I^2 = 77.6\%$). When analyzed according to the TSH suppression target, stringent suppression, but not moderate suppression, showed negative effects on postoperative BMD (SMD, -0.55; 95% CI, -0.99 to -0.10; $I^2 = 75.8\%$) (Figure 1A).

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Subgroup analysis showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (Figure 2A). Furthermore, the effects of TSH suppression on BMD were inconsistent between subgroups according to extent of thyroidectomy and baseline BMI. Detrimental effects of TSH suppression were more pronounced in patients undergoing total thyroidectomy and in those with a baseline BMI <25 kg/m^2 than in those underdoing near-total thyroidectomy and in those with a baseline BMI ≥25 kg/m^2.

Effects of TSH suppression on the femoral neck

Among the included studies, 10 studies[26,29-31,34-37,40,41] including 282 postmenopausal women and 452 controls investigated the effects of TSH suppression after thyroidectomy for thyroid cancer on the BMD of the femoral neck. The BMD of the femoral neck was similar in TSH-suppressed postmenopausal women undergoing thyroidectomy for thyroid cancer compared to that in controls (SMD, -0.32; 95% CI, -0.74 to 0.10; I^2 = 81.8%; Figure 1B). This observation was similar across subgroups according to the suppressed TSH levels. Subgroup analysis assessing the BMD of the femoral neck showed detrimental effects of TSH suppression after total thyroidectomy (SMD, -1.47; 95% CI, -2.61 to -0.33; I^2 = 90.4), but after near-total thyroidectomy (Figure 2B). Follow-up duration, race, and BMI did not affect the association between TSH suppression and the BMD of the femoral neck in postmenopausal patients undergoing thyroidectomy for thyroid cancer.

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DISCUSSION

The potential effect of exogenous hyperthyroidism on postoperative bone health in patients undergoing thyroidectomy for thyroid cancer remains a concern in terms of their long-term care. Identifying the groups that are at a risk of deteriorating bone health after thyroidectomy is crucial for tailoring TSH suppression and guiding clinical treatment in a more focused and individualized direction. In this meta-analysis of 16 studies including 1,127 participants, we identified stringent TSH suppression after thyroidectomy for thyroid cancer adversely affects the postoperative BMD in postmenopausal women. BMD is widely used for osteoporosis screening, and change in BMD is an independent risk factor for fracture.[42] Our findings suggest that the oncological benefits of stringent TSH suppression in postmenopausal women should be weighed against its possible adverse effects on postoperative bone health.

Clinical guidelines recommend the administration of levothyroxine as the optimal treatment for patients undergoing thyroidectomy for thyroid cancer; patients with residual carcinoma or those at a high risk of cancer recurrence, those at an intermediate risk of cancer recurrence, and those at a low risk of cancer recurrence should maintain TSH levels at <0.1, 0.1–0.5, and 0.5–2 mIU/L, respectively.[1,11,12] Owing to its beneficial effects on cancer recurrence[43] and overall survival,[44] stringent TSH suppression is recommended for patients at a high risk of cancer recurrence; however, there is no consensus regarding serum TSH levels below the lower limit of the normal range.[45] Because of the favorable prognosis associated with TSH suppression, patients who were treated from thyroid cancer receive life-long thyroxine treatment and have suppressed TSH levels for extended periods of time. Some studies have reported that long-term thyroxine therapy leads to an increased risk of bone fracture.[46, 47] However, the effects of TSH suppression on bone health of patients

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undergoing thyroidectomy for thyroid cancer are rarely discussed in the current guidelines, and there is a lack of studies evaluating the adverse effects according to stratified TSH levels. Several recent population-based studies have reported associations between levothyroxine doses and bone health [13, 48]. One study concluded that among patients with thyroid cancer, those who received levothyroxine had an increased risk of osteoporosis in a cumulative dose-dependent manner compared to those who did not receive levothyroxine (hazard ratio [HR] at a cumulative levothyroxine dose >395 mg, 3.62; 95% CI, 2.16 to 6.06; HR at a cumulative levothyroxine dose ≤265 mg, 1.53; 95% CI, 0.91 to 2.57).[48] In another study, 185,956 patients with thyroid cancer administered a levothyroxine dosage of ≥170 μg/day had higher risk of osteoporotic fracture (HR, 1.25; 95% CI, 1.07 to 1.45) than matched controls without cancer.[13] However, as the intensity of TSH suppression therapy was stratified based on prescribed levothyroxine doses, results from these studies reflect indirect effects of TSH concentration on bone.

In previous studies, TSH suppression after thyroidectomy significantly reduced bone mass in postmenopausal women, whereas it did not affect premenopausal women and men.[14,17, 49] As menopause and subsequent decrease in estrogen are well-known risk factors contributing to osteoporosis,[50] prominent effects of TSH suppression on bone health in postmenopausal women are clinically significant and affected subjects need to be properly managed. We observed deleterious effects of stringent TSH suppression on the BMD of the lumbar spine; however, moderate TSH suppression did not significantly affect the postoperative BMD. Inhibition of receptor activator of nuclear factor-kB ligand-induced osteoclast formation[51] and activation of Wnt-5a- and β-arrestin 1-mediated osteoblast differentiation[49, 50] support that lower TSH levels may be responsible for lowering of BMD.

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Our findings imply the need for further investigation regarding benefits of TSH suppression therapy with reference to cancer recurrence or progression and the risk of osteoporosis or fracture, especially in patients at a high risk of cancer recurrence receiving stringent TSH suppression therapy. A study including patients at a low-to-intermediate risk of cancer recurrence, defined by the American Thyroid Association,[53] demonstrated that maintaining a serum TSH level of <0.4 mU/L increased the risk of postoperative osteoporosis, but the incidence of tumor recurrence remained unchanged.[3] Further studies, preferentially with a prospective design, are required to stratify patients based on the risk of tumor recurrence and evaluate the risks and benefits of TSH suppression therapy. We agree with a review article that proposes adjusting TSH suppression strategy by evaluating the individual risk of levothyroxine therapy related to bone health in addition to the risk of cancer recurrence.[45]

The detrimental effects of TSH suppression were reported to be more prominent on the BMD of the lumbar spine than on that of the femoral neck (Figure 1). This result is consistent with that of previous studies. First, the lumbar spine is predominantly a trabecular bone; its microarchitecture is more vulnerable to osteoporotic changes than that of the cortical bone, which composes a large portion of the femoral neck.[54] Second, the mineral contents of the lumbar spine change primarily during menopause, while those of the femoral neck change at a constant rate over the lifetime.[55] In postmenopausal women, these factors may affect the interaction between the BMDs of the lumbar spine and the femoral neck and the levels of TSH and estrogen. Low BMD of the lumbar spine has clinical significance as the spine is the most common site of osteoporotic fracture[56] and spinal fractures are related to increased mortality.[57] A recent population-based study assessing the risk of fracture after

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thyroidectomy by tracing patients for 40 years also reported a significant increase in the risk of vertebral fractures (standardized incidence ratio, 2.8; 95% CI, 2.3 to 3.3) and a relatively small increase in the risk of proximal femur fractures (standardized incidence ratio, 1.3; 95% CI, 1.0 to 1.8) after thyroidectomy.[58]

The deleterious effects of TSH suppression on BMD of the lumbar spine were more prominent in patients undergoing total (vs. near-total) thyroidectomy, in Asian (vs. non-Asian) populations, and in those with BMI <25 kg/m^2 (vs. BMI ≥25 kg/m^2). Near-total thyroidectomy, which leaves a small amount of thyroid tissue (<1.0 mL), can be performed in lieu of total thyroidectomy, whereas partial or subtotal thyroidectomy is not appropriate for a thyroid malignancy.[1] Limited information is available regarding the differential effects of total and near-total thyroidectomy on bone health and BMD, and our results suggest a need for further research regarding this aspect. A study found that patients with a lower BMI undergoing thyroidectomy because of benign pathology required a significantly higher levothyroxine dose per kilogram to achieve a euthyroid state compared to those with a higher BMI (BMI <25 kg/m^2, 1.84 ± 0.07 μg/kg/day; BMI ≥35 kg/m^2, 1.39 ± 0.05 μg/kg/day); this association between BMI and the euthyroid dose was age independent.[59] We suggest levothyroxine dose adjustment and thorough monitoring of BMD in nonobese patients undergoing thyroidectomy for thyroid cancer, followed by TSH suppression.

There are several limitations in our study. First, most of the included studies had a retrospective design and the number of controlled confounding factors were heterogeneous in each study. Furthermore, studies may have been misclassified because of various factors known to affect TSH levels, such as factors related to the postmenopausal state[60] and parathyroid hormone levels.[61] Second, most of the included studies that involved TSH

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suppression reported the TSH level at a single time point. Although each study periodically assessed TSH levels to confirm maintenance within the intended level, fluctuation in the TSH levels cannot be ruled out, and consequently, the effects of these fluctuations on BMD may represent potential confounders. Third, few studies considered genetic bone susceptibility or the effects of lifestyle such as physical activity, smoking, and alcohol consumption on BMD. Fourth, some studies may have been misclassified because of the use of a first-generation TSH assay with insufficient sensitivity.[62]

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CONCLUSIONS

Stringent TSH suppression (TSH, <0.10 mIU/L) after thyroidectomy for thyroid cancer exerted detrimental effects on the postoperative BMD of the lumbar spine in postmenopausal women. Further well-designed prospective studies are needed to evaluate the change in the BMD of different bones and the risk of bone fracture according to TSH suppression levels in postmenopausal women who have undergone thyroidectomy for thyroid cancer to establish adequate post-operative management.

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COMPETING INTERESTS

The authors declare that there are no competing interests.

AUTHORS' CONTRIBUTIONS

D.K. designed the study, collected, extracted, and analyzed data. J.H. extracted and analyze data and wrote the manuscript. Y.W. interpreted the results and critically revised the manuscript. J.W.P. interpreted data and critically revised the manuscript. Y.K. designed the study, collected data, and wrote the manuscript. S.P. supervised the study and critically revised the manuscript.

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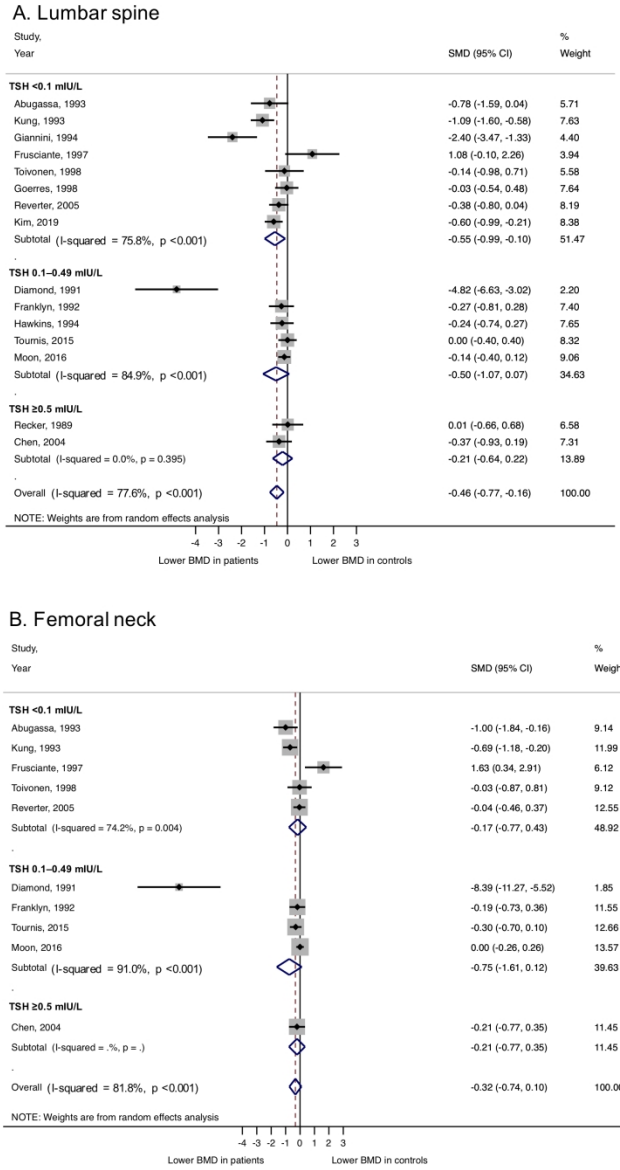
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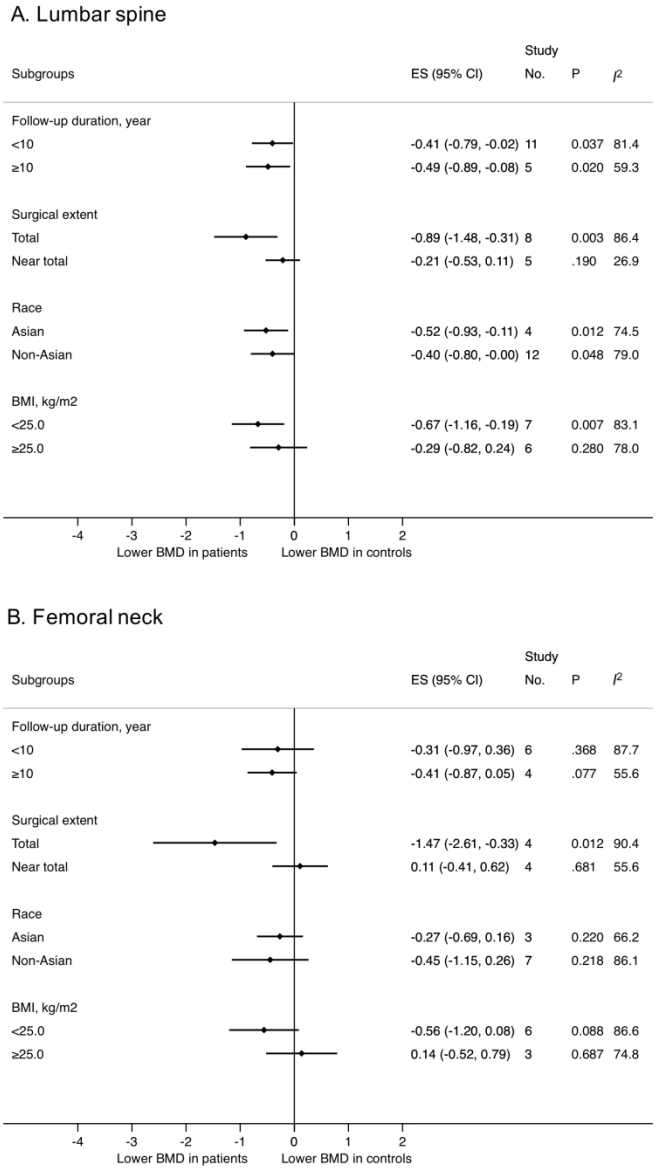
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Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women undergoing TSH suppression after thyroidectomy for thyroid cancer.
Meta-analysis results of case-control studies included in this review were presented according to TSH level. SMDs were calculated such that positive values indicated benefit for BMD in thyroidectomized group compared to controls without thyroid cancer.
TSH, thyroid-stimulating hormone; SMD, standardized mean difference; BMD, bone mineral density

172x295mm (300 x 300 DPI)



Subgroup analysis

Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women after thyroidectomy for thyroid cancer and subsequent TSH suppression by subgroups.

ES, effect size; BMD, bone mineral density; BMI, body mass index

161x278mm (300 x 300 DPI)

Supplemental Table 1. Search strategy

Database	Search strategy
PubMed	#1. thyroidectomy[mesh] #2. Thyroidectomy[tiab] OR Thyroidectomies[tiab] OR strumectomy[tiab] #3. (thyroid[mesh] OR thyroid[tiab]) AND (resection[tiab] OR surgery[tw] OR removal[tiab] OR operation[tiab] OR remove[tiab]) #4. (#1 OR #2 OR #3) #5. osteoporosis[mesh] #6. Osteoporosis[tiab] OR Osteoporoses[tiab] OR Osteoporotic[tiab] OR "Female Athlete Triad"[tiab] OR "Bone Loss"[tiab] OR "Bone Losses"[tiab] #7. (Osteopenia[tiab] OR Osteopenias[tiab]) #8. Bone Density[mesh] #9. "Bone density"[tiab] OR "Bone densities"[tiab] OR "mineral density"[tiab] OR "mineral densities"[tiab] OR "osseous density"[tiab] OR "osseous densities"[tiab] OR "bone mineral content"[tiab] OR "bone mineral contents"[tiab] OR (BMD[tiab] AND bone[tiab]) #10. bone fractures[mesh] #11. fracture[tiab] OR fractures[tiab] OR "broken bone"[tiab] OR "broken bones"[tiab] #12. (#5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11) #13. (#4 AND #12) #14. (Prognosis/Broad[filter]) AND ((thyroidectomy OR thyroidectomies OR strumectomy OR "thyroid surgery" OR "thyroid resection" OR "thyroid removal") AND (osteoporosis OR osteopenia OR "bone density" OR fracture OR fractures)) #15. (#13 OR #14) #16. ((animals [mh:noexp] OR animal experimentation[mesh] OR animal models[mesh])) #17. animals[mesh] AND human[mesh] #18. (#16 NOT #17) #19. (#15 NOT #18) #20. (#13 OR #14) Filters: Humans #21. (#19 OR #20) #22. (#19 OR #20) Filters: English DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 421

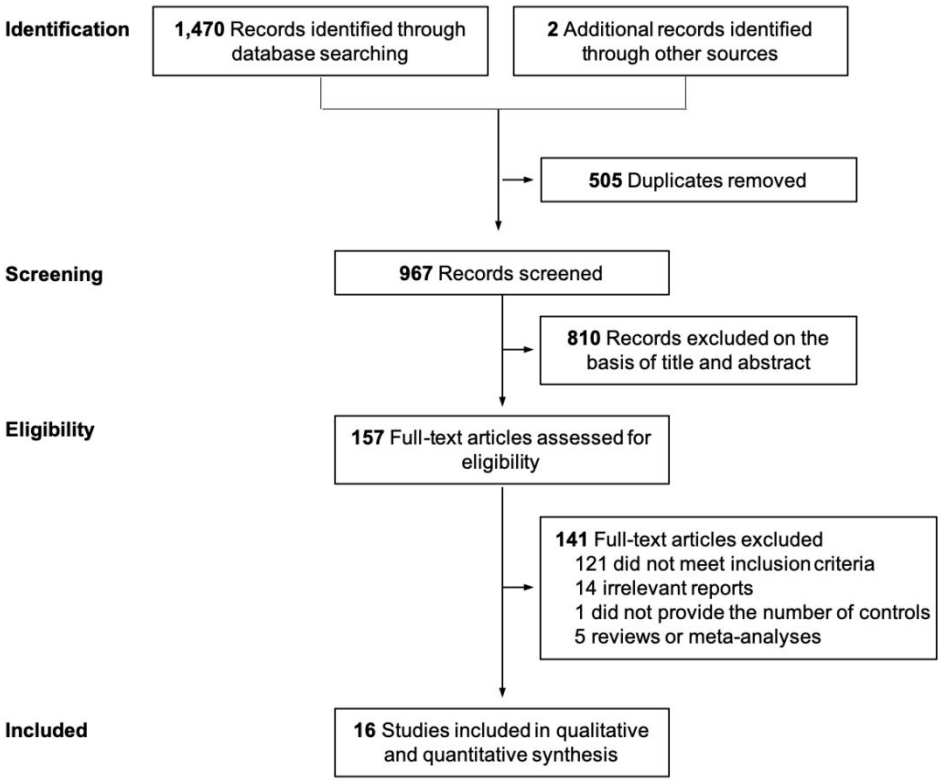
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EMBASE	#1. 'thyroid surgery'/exp #2. thyroidectomy:ti,ab OR thyroidectomies:ti,ab OR strumectomy:ti,ab OR thyroparathyroidectomy:ti,ab #3. 'thyroid gland'/de OR thyroid:ti,ab #4. resection:ti,ab,kw OR surgery:ti,ab,kw OR removal:ti,ab,kw OR operation:ti,ab,kw OR remove:ti,ab,kw #5. #3 AND #4 #6. #1 OR #2 OR #5 #7. 'osteoporosis'/exp #8. osteoporosis:ti,ab OR osteoporoses:ti,ab OR osteoporotic:ti,ab OR 'female athlete triad':ti,ab OR 'bone loss':ti,ab OR 'bone losses':ti,ab #9. 'osteopenia'/exp #10. osteopenia:ti,ab OR osteopenias:ti,ab #11. 'bone density'/exp #12. (bone NEAR/2 density) OR (bone NEAR/2 densities) OR 'mineral densit*' OR 'osseous densit*' OR 'bone mineral content*' OR (bmd:ti,ab AND bone:ti,ab) #13. 'fracture'/exp #14. fracture:ti,ab OR fractures:ti,ab OR 'broken bone*':ti,ab #15. #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 #16. #6 AND #15 #17. ('animal'/exp OR 'invertebrate'/exp OR 'animal experiment'/exp OR 'animal model'/exp OR 'animal tissue'/exp OR 'animal cell'/exp OR 'nonhuman'/exp) NOT ('human'/de OR 'normal human'/de OR 'human cell'/de) #18. #16 NOT #17 #19. #16 AND [humans]/lim #20. #18 OR #19 #21. (#18 OR #19) AND ([english]/lim) DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 1,010
Cochrane Library	#1. [mh thyroidectomy] #2. (Thyroidectomy or Thyroidectomies or strumectomy):ti,ab #3. [mh thyroid] or thyroid:ti,ab #4. (resection or surgery or removal or operation or remove):ti,ab,kw #5. #3 and #4

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	#6. #1 or #2 or #5 #7. [mh osteoporosis] #8. (Osteoporosis or Osteoporoses or Osteoporotic or "Female Athlete Triad" or "Bone Loss" or "Bone Losses"):ti,ab #9. (Osteopenia or Osteopenias):ti,ab #10. [mh "Bone Density"] #11. ("Bone density" or "Bone densities" or "mineral density" or "mineral densities" or "osseous density" or "osseous densities" or "bone mineral content" or "bone mineral contents" or (BMD and bone)):ti,ab #12. [mh "bone fractures"] #13. (fracture or fractures or "broken bone" or "broken bones"):ti,ab #14. {or #7-#13} #15. #6 and #14 DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 39
Related article search	Related articles searched according to the following references: 1) Quan ML, Pasieka JL, Rorstad O. Bone mineral density in well-differentiated thyroid cancer patients treated with suppressive thyroxine: a systematic overview of the literature. *J Surg Oncol.* 2002;79(1):62-69; discussion 69-70. 2) Heemstra KA, Hamdy NA, Romijn JA, Smit JW. The effects of thyrotropin-suppressive therapy on bone metabolism in patients with well-differentiated thyroid carcinoma. *Thyroid.* 2006;16(6):583-591. 3) Parker WA, Edafe O, Balasubramanian SP. Long-term treatment-related morbidity in differentiated thyroid cancer: a systematic review of the literature. *Pragmat Obs Res.* 2017;8:57-67. 4) Uzzan B, Campos J, Cucherat M, Nony P, Boissel JP, Perret GY. Effects on bone mass of long term treatment with thyroid hormones: a meta-analysis. *J Clin Endocrinol Metab.* 1996;81(12):4278-4289. 5) Yoon B-H, Lee Y, Oh HJ, Kim SH, Lee Y-K. Influence of thyroid-stimulating hormone suppression therapy on bone mineral density in patients with differentiated thyroid cancer: a meta-analysis DATE OF SEARCH FOR RELATED ARTICLES: April 20, 2020 TOTAL: 332 NUMBER OF RESULTS: 2

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Supplemental Figure 1. Flow diagram of reference selection process

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Supplemental Table 2. Additional information on the included studies

First author (year)	Postoperative levothyroxine dose, mean	TSH level at follow-up period, mean	Lists of exclusion criteria for medications*	Lists of exclusion criteria for medical conditions†
Recker (1989)	Range, 0.15-0.20 mg/day	Patients, 1.8 (SD, 0.9) mIU/L; Controls, 3.2 (SD, 1.6) mIU/L	Estrogens, corticosteroids, calcium	Metastatic thyroid cancer, diabetes, post-oophorectomy, post-hysterectomy
Diamond (1991)	Cumulative dose of levothyroxine, mg: 337 (SE, 72)	‡ Patients, 0.3 (SE, 0.1) mIU/L; Controls, 1.2 (SE, 0.2) mIU/L	Estrogens, calcium, vitamin D	(Stated without the list of medical conditions)
Franklyn (1992)	175 (range, 100-200) µg/day	Patients, 0.26 (SD, 0.54) mIU/L; Controls, 1.94 (SD, 1.32) mIU/L	Estrogens, calcium, vitamin D, tamoxifen, thiazide diuretics	Recurrent or metastatic thyroid cancer, osteoporotic fracture, rheumatoid arthritis, diabetes, hypoparathyroidism, alcohol abuse, chronic amenorrhea, late menarche, early menopause, post-oophorectomy
Abugassa (1993)	0.23 (SE, 0.02) mg/day	<0.1 mIU/L	Estrogens	Osteoporotic fracture, endocrinological (apart from hyperparathyroidism), hepatic, intestinal, or renal abnormalities
Kung (1993)	179 (SD, 60) µg/day	Patients, <0.05 mIU/L; Controls, 2.2 (SD, 0.8) mIU/L	Calcium, hormonal contraceptive agents	Recurrent thyroid cancer, late menopause, early menarche, post-oophorectomy, history of other thyroid disorder, family history of osteoporosis
Giannini (1994)	144.2 (SE, 4.15) µg/day	<0.1 mIU/L	(Stated without the list of drugs)	Metastatic thyroid cancer, hypoparathyroidism
Hawkins (1994)	158.3 (SD, 43.7) µg/day	0.3 (SD, 0.4) mIU/L	Estrogens, calcium, vitamin D, thiazide diuretics	(Stated without the list of medical conditions)
Frusciante (1998)	133.6 (SE, 8.4) µg/day	0.04 (SE, 0.016) mIU/L	(Stated without the list of drugs)	(Stated without the list of medical conditions)
Toivonen (1998)	215 (SD, 53) µg/day	Patients, <0.05 mIU/L; Controls, 2.4 (range, 0.06-4.1)	(Stated without the list of drugs)	(Stated without the list of medical conditions)

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		mIU/L		
Goerres (1998)	NR ††	0.019 (SD, 0.056) mIU/L	Calcitonin, calcium, vitamin D compounds	NR
Chen (2004)	Starting dose: young patients, 2.0 µg/kg/day; elderly patients, 1.5 µg/kg/day **	1.76 (SD, 0.41) mIU/L	Hormone replacement therapy, corticosteroids, tamoxifen, anticonvulsants, heparin, lithium	Metastatic or recurrent thyroid cancer, hyperparathyroidism, hypoparathyroidism, gonadal insufficiency, diabetes, Cushing's syndrome, severe liver or kidney disorder, chronic obstructive pulmonary disease, rheumatoid arthritis.
Reverter (2005)	NR	Patients, 0.04 (SD, 0.03) mIU/L; Controls, 1.8 (SD, 1.0) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, bisphosphonates, raloxifene, tamoxifen, thiazide diuretics,	Prolonged immobilization, recent bone fracture, inflammatory osteoarticular disease, diabetes, serum creatinine levels >1.3 mg/dL, increased alkaline phosphatase
Eftekhari (2008)	168.33 (SD, 24.3) µg/day	< 0.3 mIU/L	Estrogens, corticosteroids, calcium, vitamin D, bisphosphonates, tamoxifen, thiazide diuretics, anticonvulsants, heparin,	< 45 years, osteoporotic fracture, diabetes, alcohol abuse, rheumatoid arthritis, chronic amenorrhea (> 3 months), late menarche, early menopause, post oophorectomy
Tournis (2015)	122.9 (SD, 21.8) µg/day	Patients, 0.11 (SD, 0.2) mIU/L; Controls, 2.28 (SD, 1.2) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, vitamin D analogues, bisphosphonates, SERMS, denosumab, teriparatide, strontium ranelate, oral contraceptives, diuretics, lithium	Liver or renal disease, prior hyperthyroidism, hyperparathyroidism, hypoparathyroidism, malabsorption syndrome and rheumatic diseases, previous osteoporotic fracture
Moon (2016)	133.5 (SD, 33.5) µg/day	0.09 (SD, 0.13) mIU/L	Bisphosphonate, oral contraceptives, menopausal hormone therapy, SERMS, diuretics, lithium, corticosteroids	Liver or renal diseases, hyperthyroidism, hyper- or hypoparathyroidism, malabsorption syndrome, rheumatic diseases
Kim (2019)	135 (SD, 3.06) µg/day	Patients, 0.000 (IQR 0.00 to 0.242); Controls, 2.450 (IQR 1.263 to 3.813)	Glucocorticoids, anti-osteoporotic drugs	Renal or hepatic impairment

TSH = thyroid stimulating hormone; NR = not reported; NA = not applicable; SD = standard deviation; SE = standard error; SERMS = selective estrogen receptor modulators; hypoPT = patients with hypoparathyroidism; DTC = differentiated thyroid carcinoma; IQR = interquartile range; NTG = non-toxic goiter

* Sixteen studies presented the list of excluded medications, and 6 studies stated that patients who had taken any medication influencing bone density were excluded from the study.

† Sixteen studies presented the list of excluded comorbidities, and 6 studies stated that patients who had any medical condition influencing bone density were excluded from the study.

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‡ Lower limit of detection for TSH in this assay was 0.3 mIU/L.

§ Lower limit of detection for TSH in this assay was 0.15 mIU/L.

‖ Among 50 participants, 12 were treated with dihydrotachysterol and another 2 with calcitriol, besides calcium tablets. Two patients had well-controlled type 2 diabetes. Three women had been taking hormonal contraceptives for 1 or 3 years, and another had a received low-dose post-menopausal estrogen substitution.

¶ Lower limit of detection for TSH in this study was 0.2 mIU/L.

†† Mean cumulated T4 dose of patients was reported as 8527.3 (6102.6) μg/kg body weight.

** The dose was adjusted at an increment/decrement of 12.5 to 25 μg/day according to the TSH value at 8- to 12-week intervals.

‡‡ Levothyroxine dose was adjusted to keep the TSH level under or equal to 0.1 mIU/L.

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	#1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	#2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	#4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	#4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	#5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	#5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	#5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplemental Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Supplemental Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	#6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	#6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	#6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	#6-7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	#7

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	#7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	#7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Supplemental Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	#8-13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	#10-13
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Figure 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Figure 1
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Figure 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	#15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	#18-19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	#20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	#21

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal females: A systematic review and meta-analysis

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Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal females: A systematic review and meta-analysis

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Abstract

Objectives: We assessed thyroid stimulating hormone (TSH) suppression effects on bone mineral density (BMD) in postmenopausal females who underwent thyroidectomy.

Data sources: PubMed, EMBASE, Cochrane Library, Web of Science, and SCOPUS were searched from inception to February 24, 2021.

Study selection: Case control studies were included.

Data extraction and synthesis: Two authors independently reviewed the studies, extracted the data, and performed meta-analysis of eligible studies.

Research design and methods: Studies evaluating BMD in postmenopausal females with thyroid cancer who received thyroidectomy and levothyroxine administration were included. Differences in BMD were presented as standardized mean differences (SMDs). Meta-analyses were conducted using a random-effects model.

Results: Analysis of 16 case control studies (426 patients and 701 controls without thyroid cancer) showed that stringent TSH suppression (TSH, <0.10 mIU/L) after thyroidectomy had deleterious effects on the BMD of the lumbar spine in postmenopausal females compared to controls (SMD, -0.55; 95% confidence interval [CI], -0.99 to -0.10; $I^2 = 75.8\%$). There was no significant difference in patients with moderate TSH suppression (TSH, 0.10–0.49 mIU/L). TSH suppression in postmenopausal females was not significantly associated with lower femoral neck BMD. Subgroup analysis of the lumbar spine showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (SMD, -0.32; 95% CI, -0.50 to -0.14). Subgroup analysis of the femoral neck showed that total thyroidectomy was related to detrimental effects on the BMD of the femoral neck (SMD, -0.60;

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95% CI, -0.89 to -0.31; I2 = 90.4%), but near-total thyroidectomy was not (SMD, 0.00; 95% CI, -0.30 to 0.30; I2 = 55.6%).

Conclusions: Stringent TSH suppression had deleterious effects on the BMD of the lumbar spine after thyroidectomy in postmenopausal females. Further studies are needed to determine whether stringent TSH suppression after thyroidectomy increases the fracture risk.

PROSPERO registration: CRD42018104282

Keywords: thyroid cancer, thyroidectomy, bone mineral density, thyroid stimulating hormone, thyroid stimulating hormone suppression

Strengths and limitations of this study

- This is the first systematic review and meta-analysis to qualitatively collate and assess the results of studies that investigated the effect of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal females.
- We investigated the association between thyroid stimulating hormone suppression and lumbar spine or femoral neck bone mineral density in different subgroups (follow-up duration, surgical extent, ethnicity, and body mass index).
- A possible limitation is that only a few prospective studies were included, and the effects of notable confounding factors and possible fluctuations in the postoperative thyroid stimulating hormone levels were not reflected in the analysis.

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INTRODUCTION

Thyroid stimulating hormone (TSH) levels are suppressed after thyroidectomy for thyroid cancer.[1-3] It is necessary to assess the risk-benefit and adverse effects of TSH suppression because patients with thyroid cancer have a favorable prognosis.[4] To prevent cancer recurrence and reduce cancer-related mortality, levothyroxine is used to achieve TSH suppression,[5] as it inhibits the production of circulating TSH, a factor known to stimulate the proliferation of thyrocytes.[6] However, supraphysiological doses of exogenous levothyroxine may cause side effects, particularly regarding bone health.[2, 7] Endogenous hyperthyroidism increases the risk of osteoporosis and osteoporotic fractures.[8, 9] However, the effects of TSH suppression due to levothyroxine administration after thyroidectomy are controversial.[10] Some studies reported a lack of deleterious effects on the bone mineral density (BMD) of long-term thyroxine suppressive therapy for differentiated thyroid carcinoma.[11-14] In contrast, other studies reported that thyroxine suppressive therapy had a detrimental effect on BMD.[15, 16]

Although different levels and durations of TSH suppression are recommended post-thyroidectomy depending on the risk of thyroid cancer recurrence,[1, 17, 18] the impact of varying levels of TSH suppression on bone health is unclear. Several studies have evaluated the association between TSH suppression levels and fractures.[8, 19, 20] One of them reported that females with a low TSH level (<0.1 mU/L) had a threefold increased risk for hip fractures and a fourfold increased risk for vertebral fractures compared with females who had normal TSH levels.[20] However, this study focused on the risk of fracture rather than BMD in females with low serum TSH levels. Several literature reviews have reported that suppressed TSH levels lead to osteoporosis and decreased bone mineral density (BMD) after thyroidectomy for thyroid cancer in postmenopausal

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females.[21-26] However, no further studies have investigated the effects of varying TSH levels on BMD.

In this study, we performed a systematic review and meta-analysis of the current literature to investigate the effects of different levels of TSH suppression after thyroidectomy on postoperative BMD in postmenopausal females with thyroid cancer.

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MATERIALS AND METHODS

An expert panel was formed to define the scope of the meta-analysis, literature search, eligibility criteria, and statistical methods according to the preferred reporting items for systematic reviews and meta-analysis guidelines.[27] This systematic review and meta-analysis protocol was registered in the PROSPERO database (registration number, CRD42018104282). At the time of initial registration of PROSPERO, patients who underwent thyroidectomy due to any thyroid disease were registered. However, there are few studies on BMD after thyroidectomy in typical hyperthyroidism diseases such as Graves' disease, toxic multinodular goiter, and toxic adenoma, excluding cancer, so our study was limited to only patients with thyroid cancer.

Patient and public involvement statement

This meta-analysis was based on study-level data, and no individual-level data were involved in the study or in defining the research question or outcome measures.

Ethics approval statement

This study was approved by the Institutional Review Board of Korea University Anam Hospital (registration number: 2021AN0165).

Literature search

A trained librarian searched the PubMed, EMBASE, Cochrane Library, Web of Science, and SCOPUS databases to identify relevant studies published before February 24, 2021 (Supplemental Table 1). The following search terms, adapted to each database, were used: "*thyroidectomy*," "*levothyroxine*," "*thyroid stimulating hormone*," "*bone density*,"

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"*osteoporosis*," and "*bone fractures*." We also included studies reported in earlier systematic reviews,[21-24, 26] and those identified from the bibliography of the included articles.

Eligibility criteria and study selection

The inclusion criteria were as follows: (1) all participants underwent thyroidectomy for thyroid cancer, followed by the administration of levothyroxine; (2) availability of BMD data, irrespective of whether this was the primary endpoint; (3) availability of data on the lumbar spine and femoral neck; (4) comparison of postoperative outcomes between the study and control groups (matched healthy postmenopausal females were enrolled); and (5) the most recent or informative publication data, in the case of multiple publications from the same population. The exclusion criteria were as follows: (1) only abstract available; (2) case reports, reviews, and editorials; (3) studies with pediatric or pregnant populations; (4) publication language other than English; and (5) missing measures of variability (standard deviations, standard errors, or interquartile ranges) when effect estimates were presented.

Two authors (D.K. and Y.K.) independently assessed the eligibility of all studies identified through an electronic literature search based on the titles and abstracts. Eligible articles with full texts were further evaluated for inclusion. Disagreements were resolved by consensus.

Data extraction and risk-of-bias assessment

A standard template suggested by Cochrane was adapted and applied for data extraction.[28] Two authors (D.K. and J.H.) independently extracted the data, including the first author's name; year of publication; study location; study design; age, body mass index (BMI), ethnicity, number of the study participants; etiology; extent of thyroidectomy; location and value

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of the BMD measurement; follow-up duration after thyroidectomy; levothyroxine dose; preoperative and postoperative laboratory results including TSH, free thyroxine, parathyroid hormone, and calcium levels; medications used by the study participants; and comorbidities.

The quality of the included studies was assessed using the Newcastle-Ottawa Quality Assessment Scale.[29] This assesses the quality of nonrandomized studies according to three variables: selection of the study population, comparability, and exposure factor.

Data synthesis and analysis

Each outcome was presented with estimates calculated as standardized mean differences (SMDs). Assuming that the baseline values of patients and controls without thyroid cancer were balanced and the confounders were controlled by the study design, we compared the measurements at the end of the follow-up period. The SMDs were coded so that positive values represented a higher BMD in the thyroidectomy group than in the control group, with 0.2, 0.5, and 0.8, corresponding to small, moderate, and large effects, respectively.[30] We estimated the pooled SMDs and 95% confidence intervals (CIs) using the random-effects model (DerSimonian and Laird method). Statistical significance was set at a two-tailed p-value of <0.05. We assessed heterogeneity using the Mantel–Haenszel model and *I2* values (the percentage of variance in the pooled estimate due to between-study differences), with *I2* thresholds of ≤25% (low), 26%–50% (moderate), and >50% (high).

Subgroup analyses were performed considering the TSH level (categorized as stringent suppression, <0.10 mIU/L; moderate suppression, 0.10–0.49 mIU/L; and lower normal, 0.50–2.0 mIU/L). Other investigated factors included the follow-up duration, baseline age, baseline BMI, ethnicity, and extent of thyroidectomy (total vs. near-total). Publication bias was assessed based

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on funnel plot asymmetry using Begg's and Egger's tests at a p-value of < 0.1. Stata version 13.1 software (StataCorp., College Station, TX, USA) was used for all statistical analyses.

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RESULTS

Studies included in the meta-analysis

A systematic literature search retrieved a total of 3,063 studies (Supplemental Figure 1). We included two studies from the bibliography of the included articles. Of the included studies, 1,305 duplicate studies were excluded. The remaining 1,760 articles were screened based on the title and abstract and assessed for eligibility. The full text of 160 potentially eligible studies was evaluated by two independent authors (D.K. and J.H.), of which 144 studies did not meet the inclusion criteria (reasons for exclusion are summarized in Supplemental Figure 1). Sixteen studies,[11-16, 31-40] including 1,127 postmenopausal females, met the inclusion criteria. Table 1 lists the summary and characteristics of each study.

TABLE 1. Characteristics of the studies included in the analysis

First author (year, location)	Study population				Methods		Outcomes
	Age, mean (SD), years	Etiology	Extent of thyroidectomy	No. of postmenopausal participants, n (total=1,127)	Assessment tool of bone density (measurement site included in the meta-analysis)	Outcomes included in the meta-analysis (unit)	Assessment time after thyroidectomy, mean, year
Recker (1989, USA)[31]	50.2	DTC	Total	Patients, 18; Controls, 16	DPA (lumbar spine and	BMD (g/m^2)	9.6

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					femoral regions)		
Diamond (1991, Australia)[16]	59 (2.8)	DTC	Total	Patients, 10; Controls, 10	DPA (Lumbar spine, femoral neck)	BMD (g/m^2)	5.9
Franklyn (1992, UK)[32]	63.4 (8.1)	DTC	Subtotal[d]	Patients, 26; Controls, 26	DEXA (lumbar spine and femoral regions)	BMD (g/m^2)	NR
Abugassa (1993, Sweden)[33]	58 (2)	Thyroid cancer	Total	Patients, 13; Controls, 12	DPA (lumbar spine and femoral regions)	BMD (g/cm^2)	13.0
Kung (1993, China)[15]	62 (8)	Thyroid cancer	Total	Patient, 34; control, 34	DEXA (lumbar spine, femoral regions)	BMD (g/cm^2)	12.2
Giannini (1994, Italy)[34]	Patients, 57.6 (1.7); Contro	Non-medullary thyroid	Total	Patients, 13; Controls, 11	DEXA (lumbar spine,	BMD (g/cm^2)	7.6

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	l, 56 (0.4)	carcinoma[c]			femoral regions)		
Hawkins (1994, Spain)[35]	Patients, 59.6 (7.5); Control, 49.3 (5.2)	Thyroid cancer	Near-total	Patients, 21; Controls, 53	DEXA (lumbar spine)	BMD (g/cm^2)	6.2
Frusciante (1998, Italy)[11]	Patients, 55.3 (1.7); Controls, 53.3 (2.5)	DTC	Near-total	Patients, 6; Controls, 7	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.7
Toivonen (1998, Finland)[12]	Patients, 45 (range, 27 to 71); Controls, 43 (range, 26 to 65)[a]	DTC	Near-total	Patients, 10; Controls, 12	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	10
Goerres (1998,	Patients, 61.6	Thyroid cancer	Total	Patients, 23; Controls, 42	DEXA (lumbar	BMD (g/cm^2)	8.8

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Switzerland)[14]	(12.0); Controls, 562.7 (8.1)				spine, femoral neck)		
Chen (2004, China)[36]	57.7 (6.9)	DTC	Near-total	Patients, 25; Controls, 25	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	7.8
Reverter (2005, Spain)[13]	58 (9)	DTC	Near-total	Patients, 44; Controls, 44	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	12.0
Eftekhari (2008, Iran)[37]	Patients, 51.7 (7.3); Controls, 51.2 (7.8) [a]	DTC	NR	Patients, 33; Controls, 33	DEXA (lumbar spine)	BMD (g/cm^2)	1.2
Tournis (2015, Greece)[38]	Patients, 56.7 (5.9); Contro	DTC	Total	Patients, 40; Controls, 60	DEXA (lumbar spine, femoral neck)	BMD (mg/cm^3) [b]	5.0

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	ls, 57.8 (6.1)						
Moon (2016, Korea)[39]	Patients, 61.4 (7.6); Controls, 61.4 (7.6)	DTC	NR	Patients, 74; Controls, 222	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.2
Kim (2019, Korea)[40]	Patients, 53; Controls, 55	DTC	Total	Patients, 36; Controls, 94	DEXA (lumbar spine)	BMD (g/cm^2)	4.9

SD = standard deviation; NR = not reported; DPA = dual photon absorptiometry; DEXA = dual-energy x-ray absorptiometry; DTC = differentiated thyroid carcinoma; BMD = bone mineral density

[a] These studies presented the mean or median age of the total study population, including pre- and postmenopausal females.

[b] This value was calculated with the method of volumetric BMD. For more information, refer to Tournis et al. (2015).

[c] This study involved patients with non-medullary thyroid carcinoma without any kind of drug treatment known to influence bone turnover or diseases affecting calcium metabolism.

[d] This study which involved subjects that underwent subtotal thyroidectomy was excluded from the total versus near-total subgroup analysis.

Studies were published as early as 1989, and the follow-up duration after thyroidectomy ranged from one to 15 years. Three studies were prospective,[13, 16, 38] while three involved consecutive sampling (Table 2).[14, 33, 37]

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TABLE 2. Quality rating for studies included in the analysis

First author (Year)	Design	Sampling strategy	Multisite	Sample size	Method of controlling for confounders	Confounders included in the analysis (Confounders included in the exclusion criteria are presented in the right two columns)	Exclusion criteria for medication [a]	Exclusion criteria for medical conditions [b]
Recker (1989)[31]	Retrospective cohort with matched control	Nonconsecutive	No	34	Matching, Exclusion	NR	Stated with the list of drugs	Stated with the list of medical conditions
Diamond (1991)[16]	Prospective cohort with matched control	Nonconsecutive	No	48	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated
Frankly n	Retrospective cohort	Nonconsecutive	No	98	Matching, Exclusion	Matching: age, sex,	Stated with the	Stated with the

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(1992)[32]	with matched control					BMI, menopausal status, smoking history, calcium intake score	list of drugs	list of medical conditions
Abugassa (1993)[33]	Retrospective cohort with matched control	Consecutive	No	25	Matching, Exclusion	Matching: age, sex, menopausal status, postoperative time, postmenopausal age	Stated with the list of drugs	Stated with the list of medical conditions
Kung (1993)[15]	Retrospective cohort with matched control	Nonconsecutive	No	68	Matching, Exclusion	NR	Stated with the list of drugs	Stated with the list of medical conditions
Giannini (1994)[34]	Retrospective cohort with matched control	Nonconsecutive	No	46	Matching, Exclusion	Matching: sex, height, weight, menopausal status	Stated	Stated

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Hawkins (1994)[35]	Retrospective cohort with matched control	Nonconsecutive	No	74	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated
Fruscian te (1998)[11]	Retrospective cohort with matched control	Nonconsecutive	No	39	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated	Stated
Toivonen (1998)[12]	Retrospective cohort with matched control	Nonconsecutive	No	67	Matching, Exclusion	Matching: age, sex, menopausal status	Stated	Stated
Goerres (1998)[14]	Retrospective cohort with matched control	Consecutive	No	114	Matching, Exclusion	Matching: age, sex	Stated with the list of drugs	NR
Chen (2004)[36]	Retrospective cohort with matched control	Nonconsecutive	No	50	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions

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Reverter (2005)[13]	Prospective cohort with matched control	Nonconsecutive	No	88	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions
Eftekhari (2008)[37]	Retrospective cohort with matched control	Consecutive	No	132	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions
Tournis (2015)[38]	Prospective cohort with matched control	Nonconsecutive	No	169	Matching, Exclusion	Matching: age, sex, BMI, menopausal status, calcium intake, years since menopause in case of postmenopausal females	Stated with the list of drugs	Stated with the list of medical conditions

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Moon (2016)[39]	Retrospective cohort with matched control	Nonconsecutive	No	496	Matching, Exclusion	Matching: age, sex	Stated with the list of drugs	Stated with the list of medical conditions
Kim (2019)[40]	Retrospective cohort with matched control	Nonconsecutive	No	130	Matching	Matching; age, BMI	NR	NR

NR = not reported; NA = not applicable; BMI = body mass index; TSH = thyroid stimulating hormone; BMD = bone mineral density

[a] Categories include NR (not reported), stated (without the list of drugs), and stated with the list of drugs. The lists of drugs for exclusion are summarized in Supplemental Table 2.

[b] Categories include NR (not reported), stated (without the list of drugs), and stated with the list of drugs. The lists of medical conditions for exclusion are summarized in Supplemental Table 2.

[c] Additional analyses used multivariable regression to evaluate the association between the TSH level and bone density after thyroidectomy controlled for TSH, free thyroxine, urinary calcium, serum alkaline phosphatase, serum osteocalcin, serum telopeptide, BMD of the lumbar spine, and BMD of the femoral hip. The comparison of bone density between patients and controls was performed using a paired *t*-test without controlling for confounders.

All studies, except one,[14] controlled for confounders in the study design using matching factors and exclusion criteria (Table 2 and Supplemental Table 2). The quality scores are shown in Supplemental Table 3, with scores ranging from six to nine. All studies were found to be of

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moderate to high quality. After thyroidectomy, the mean TSH levels ranged between 0.00 and 1.8 mIU/L. The postoperative levothyroxine doses are summarized in Supplemental Table 2.

Effects of TSH suppression on the lumbar spine

All the included studies reported BMD of the lumbar spine in 426 postmenopausal females that underwent thyroidectomy for thyroid cancer and 701 controls. Overall, TSH suppression after thyroidectomy showed detrimental effects on the BMD of the lumbar spine (SMD, -0.46; 95% CI, -0.77 to -0.16; I^2 = 77.6%; p =.003). When analyzed according to the TSH suppression target, stringent suppression, but not moderate suppression, showed negative effects on the postoperative BMD (SMD, -0.55; 95% CI, -0.99 to -0.10; I^2 = 75.8%; p =.016) (Figure 1A). Funnel plots indicated that there was no significant publication bias with stringent TSH suppression (TSH < 0.1 mIU/L) (Supplemental Figure 2A), which was supported by Egger's test (p =0.940).

Effects of TSH suppression on the femoral neck

Ten of the studies (282 postmenopausal females and 452 controls) investigated the effects of TSH suppression after thyroidectomy for thyroid cancer on the BMD of the femoral neck.[11-13, 15, 16, 32, 33, 36, 38, 39] The BMD of the femoral neck was similar in TSH-suppressed postmenopausal females that underwent thyroidectomy for thyroid cancer compared to that in controls (SMD, -0.32; 95% CI, -0.74 to 0.10; I^2 = 81.8%; p =.133; Figure 1B). This observation was similar across subgroups according to the suppressed TSH levels. Funnel plots indicated no significant publication bias (Supplemental Figure 2B), as supported by Egger's test (p =0.177).

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Subgroup analyses

A subgroup analysis showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (Figure 2A). No significant differences in subgroups with different ethnicities and BMIs were detected. Furthermore, the detrimental effects of TSH suppression were more pronounced in patients who underwent total thyroidectomy than in those that underwent near-total thyroidectomy without statistical significance ($p = 0.065$). A significant interaction between surgical extent and BMD of the femoral neck (Figure 2B; p for interaction, 0.004) was observed. Moreover, total thyroidectomy was related to detrimental effects on the BMD of the femoral neck (SMD, -0.60; 95% CI, -0.89 to -0.31; $I^2 = 90.4\%$; $p < 0.001$), but near-total thyroidectomy was not (SMD, 0.00; 95% CI, -0.30 to 0.30; $I^2 = 55.6\%$; $p = 0.996$). The follow-up duration, ethnicity, and BMI did not affect the association between TSH suppression and femoral neck BMD in postmenopausal patients that underwent thyroidectomy for thyroid cancer.

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DISCUSSION

Summary of the findings

The potential effect of hyperthyroidism on postoperative bone health in patients that underwent thyroidectomy for thyroid cancer remains a concern in terms of long-term care. Identifying the groups that are at risk of deteriorating bone health after thyroidectomy is crucial for tailoring TSH suppression and guiding clinical treatment in a more focused and individualized direction. In this meta-analysis of 16 studies, including 1,127 participants, we identified stringent TSH suppression after thyroidectomy for thyroid cancer that adversely affects the postoperative BMD in postmenopausal females. BMD is widely used for osteoporosis screening, and BMD changes are an independent risk factor for fracture.[41] Our findings suggest that the oncological benefits of stringent TSH suppression in postmenopausal females should be weighed against its possible adverse effects on postoperative bone health.

Comparison with other studies

Clinical guidelines recommend the administration of levothyroxine as the optimal treatment for patients undergoing thyroidectomy for thyroid cancer. Patients with residual carcinoma or those at a high risk of cancer recurrence, those at an intermediate risk of cancer recurrence, and those at a low risk of cancer recurrence should maintain TSH levels at <0.1, 0.1–0.5, and 0.5–2 mIU/L, respectively.[1, 17, 18] Due to its beneficial effects on cancer recurrence,[42] and overall survival,[43] stringent TSH suppression is recommended for patients at a high risk of cancer recurrence. However, there is no consensus regarding serum TSH levels below the lower limit of the normal range.[44] Because of the favorable prognosis associated with TSH suppression, patients with thyroid cancer receive life-long thyroxine treatment and

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have suppressed TSH levels for extended periods. Some studies have reported that long-term thyroxine therapy leads to an increased risk of bone fractures.[45, 46] However, the effects of TSH suppression on the bone health of patients undergoing thyroidectomy for thyroid cancer are rarely discussed in the current guidelines, and few studies evaluate the adverse effects according to stratified TSH levels. Several recent population-based studies have reported associations between levothyroxine doses and bone health.[19, 47] One study concluded that among patients with thyroid cancer, those who received levothyroxine had an increased risk of osteoporosis in a cumulative dose-dependent manner compared to those who did not receive levothyroxine (hazard ratio [HR] at a cumulative levothyroxine dose >395 mg, 3.62; 95% CI, 2.16 to 6.06; HR at a cumulative levothyroxine dose ≤265 mg, 1.53; 95% CI, 0.91 to 2.57).[47] In another study, 185,956 patients with thyroid cancer that were administered a levothyroxine dosage of ≥170 μg/day had a higher risk of osteoporotic fractures (HR, 1.25; 95% CI, 1.07 to 1.45) than matched controls without cancer.[19] However, as the intensity of TSH suppression therapy was stratified based on prescribed levothyroxine doses, results from these studies reflect the indirect effects of TSH concentration on the bone.

According to a series of previous studies, TSH suppression after thyroidectomy significantly reduced bone mass in postmenopausal females, whereas it did not affect premenopausal females and males.[21, 24, 48] Menopause and the subsequent decrease in estrogen are well-known risk factors that contribute to osteoporosis.[49] The prominent effects of TSH suppression on bone health in postmenopausal females are clinically significant, and affected subjects need to be appropriately managed. We observed deleterious effects of stringent TSH suppression on the BMD of the lumbar spine. However, moderate TSH suppression did not

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significantly affect postoperative BMD. Inhibition of the receptor activator of nuclear factor-κB ligand-induced osteoclast formation,[50] and activation of Wnt-5a- and β-arrestin 1-mediated osteoblast differentiation support that lower TSH levels may be responsible for lowering BMD.[48, 49]

BMD comparison on the lumbar spine and femoral neck

The detrimental effects of TSH suppression have been reported to be more prominent on the BMD of the lumbar spine than on that of the femoral neck (Figure 1). This result is consistent with that of previous studies. First, the lumbar spine is predominantly a trabecular bone. Its microarchitecture is more vulnerable to osteoporotic changes than that of the cortical bone, which comprises a large portion of the femoral neck.[51] Second, the mineral contents of the lumbar spine change primarily during menopause, while those of the femoral neck change at a constant rate over one's lifetime.[52] These factors may affect the interaction between the BMD of the lumbar spine and femoral neck and TSH and estrogen levels in postmenopausal females. Low BMD of the lumbar spine is clinically significant as the spine is the most common osteoporotic fracture site,[53] and spinal fractures are related to increased mortality.[54] A recent population-based study assessing the risk of fracture after thyroidectomy by tracing patients for 40 years also reported a significant increase in the risk of vertebral fractures (standardized incidence ratio, 2.8; 95% CI, 2.3 to 3.3) and a relatively small increase in the risk of proximal femur fractures (standardized incidence ratio, 1.3; 95% CI, 1.0 to 1.8) after thyroidectomy.[55]

The deleterious effects of TSH suppression on the BMD of the lumbar spine were more prominent in patients that underwent total (vs. near-total) thyroidectomy, in Asian (vs. non-

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Asian) populations, and in those with a BMI <25 kg/m^2 (vs. BMI ≥25 kg/m^2). Near-total thyroidectomy, which leaves a small amount of thyroid tissue (<1.0 mL), can be performed instead of total thyroidectomy, whereas partial or subtotal thyroidectomy is not appropriate for thyroid malignancy.[1] Limited information is available regarding the differential effects of total and near-total thyroidectomy on bone health and BMD, suggesting the need for further research regarding this aspect. A study found that patients with a lower BMI that underwent thyroidectomy because of benign pathology required a significantly higher levothyroxine dose per kilogram to achieve a euthyroid state compared to those with a higher BMI (BMI <25 kg/m^2, 1.84 ± 0.07 μg/kg/day; BMI ≥35 kg/m^2, 1.39 ± 0.05 μg/kg/day); this association was age independent.[56] We suggest levothyroxine dose adjustment and thorough monitoring of the BMD in nonobese patients undergoing thyroidectomy for thyroid cancer, followed by TSH suppression.

Limitations

Our study has several limitations. First, most of the included studies had a retrospective design, and the number of controlled confounding factors was heterogeneous in each study. Furthermore, studies may have been misclassified because of various factors known to affect TSH levels, such as factors related to the postmenopausal state,[57] and parathyroid hormone levels.[58] Second, most of the included studies that involved TSH suppression reported the TSH level at a single time point. Although each study periodically assessed TSH levels to confirm maintenance within the intended level, fluctuations in TSH levels cannot be ruled out, and consequently, the effects of these fluctuations on BMD may represent potential confounders. Third, few studies have considered genetic bone susceptibility or the effects of lifestyle factors

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such as physical activity, smoking, and alcohol consumption on the BMD. Fourth, some studies may have been misclassified because of the use of a first-generation TSH assay with insufficient sensitivity.[59] Fifth, most of the cases were differentiated thyroid carcinoma, but thyroid cancer etiology was not reported, or patients with other etiologies were included. Therefore, additional treatments or differences in molecular pathogenesis may be the cause of heterogeneity. Sixth, most studies did not clarify whether all of the included patients were high risk or suppressed below the TSH level of 0.1 mIU/L. Therefore, statistical analyses were performed based on the average TSH levels presented in each study. Seventh, other variables that may influence hypoparathyroidism or vitamin D levels, such as autoimmune conditions, genetic variants, radiation, mineral deposition, and magnesium deficiency, were not evaluated.[60] Eighth, only case control studies were enrolled. A prospective cohort and longitudinal study are required. Ninth, funnel plots of the lumbar spine indicated that there was some publication bias with overall and moderate TSH suppression (TSH, 0.10–0.49 mIU/L) (Supplemental Figure 2a.), as shown by the Egger test (p =.098;.082). Interpretation of the overall and moderate TSH suppression groups requires further investigation. Tenth, few studies have analyzed the osteoporosis and fracture risk results according to the TSH suppression level, so this was not covered in our systematic review. Further investigation is needed regarding the benefits of TSH suppression therapy in relation to cancer recurrence or progression and the risk of osteoporosis or fractures, particularly in patients at a high risk of cancer recurrence receiving stringent TSH suppression therapy.

CONCLUSIONS

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Stringent TSH suppression (TSH level, <0.10 mIU/L) after thyroidectomy for thyroid cancer was found to exert detrimental effects on the postoperative BMD of the lumbar spine in postmenopausal females. Further well-designed prospective studies are needed to evaluate the change in the BMD of different bones and the risk of bone fractures according to the TSH suppression levels in postmenopausal females who have undergone thyroidectomy for thyroid cancer to establish adequate postoperative management.

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FIGURE LEGENDS

Figure 1. Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal female undergoing TSH suppression after thyroidectomy for thyroid cancer.

Meta-analysis results of case-control studies included in this review were presented according to TSH level. SMDs were calculated such that positive values indicated benefit for BMD in thyroidectomized group compared to controls without thyroid cancer.

TSH, thyroid-stimulating hormone; SMD, standardized mean difference; BMD, bone mineral density.

Figure 2. Subgroup analysis

Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women after thyroidectomy for thyroid cancer and subsequent TSH suppression by subgroups.

ES, effect size; BMD, bone mineral density; BMI, body mass index.

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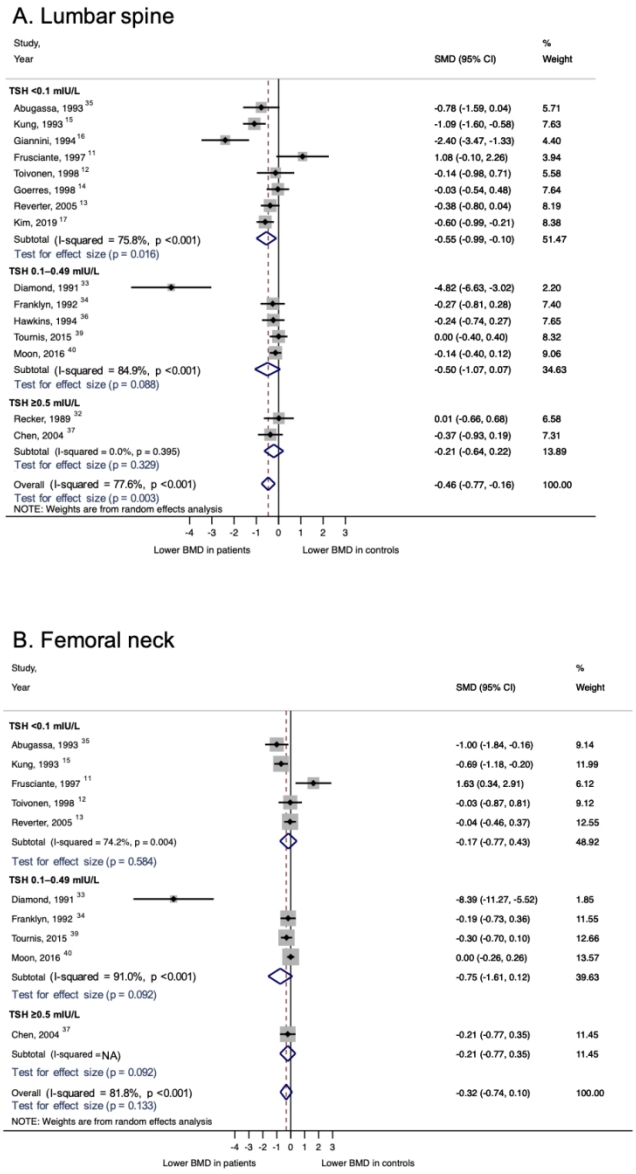


Figure 1. Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women undergoing TSH suppression after thyroidectomy for thyroid cancer. Meta-analysis results of case-control studies included in this review were presented according to TSH level. SMDs were calculated such that positive values indicated benefit for BMD in thyroidectomized group compared to controls without thyroid cancer.TSH, thyroid-stimulating hormone; SMD, standardized mean difference; BMD, bone mineral density

345x611mm (72 x 72 DPI)

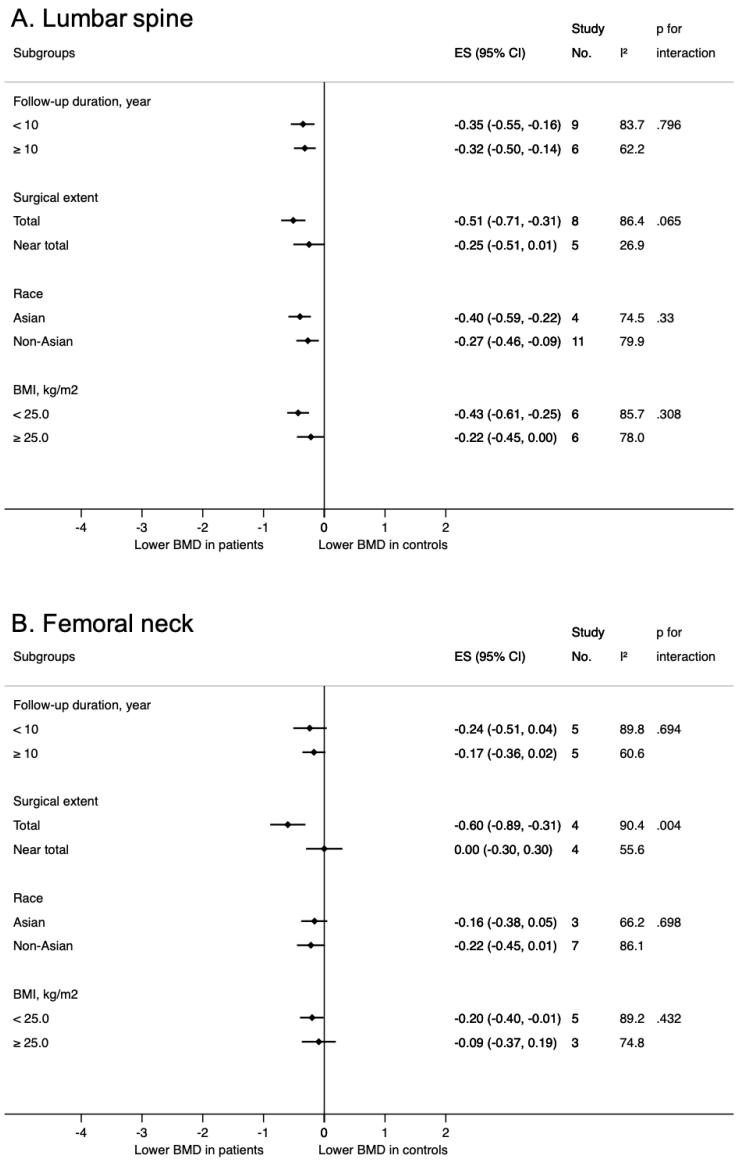


Figure 2. ubgroup analysisPostoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women after thyroidectomy for thyroid cancer and subsequent TSH suppression by subgroups.ES, effect size; BMD, bone mineral density; BMI, body mass index

352x541mm (72 x 72 DPI)

Supplemental Table 1. Search strategy

Database	Search strategy
PubMed	#1. thyroidectomy[mesh] #2. Thyroidectomy[tiab] OR Thyroidectomies[tiab] OR strumectomy[tiab] #3. (thyroid[mesh] OR thyroid[tiab]) AND (resection[tiab] OR surgery[tw] OR removal[tiab] OR operation[tiab] OR remove[tiab]) #4. (#1 OR #2 OR #3) #5. osteoporosis[mesh] #6. Osteoporosis[tiab] OR Osteoporoses[tiab] OR Osteoporotic[tiab] OR "Female Athlete Triad"[tiab] OR "Bone Loss"[tiab] OR "Bone Losses"[tiab] #7. (Osteopenia[tiab] OR Osteopenias[tiab]) #8. Bone Density[mesh] #9. "Bone density"[tiab] OR "Bone densities"[tiab] OR "mineral density"[tiab] OR "mineral densities"[tiab] OR "osseous density"[tiab] OR "osseous densities"[tiab] OR "bone mineral content"[tiab] OR "bone mineral contents"[tiab] OR (BMD[tiab] AND bone[tiab]) #10. bone fractures[mesh] #11. fracture[tiab] OR fractures[tiab] OR "broken bone"[tiab] OR "broken bones"[tiab] #12. (#5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11) #13. (#4 AND #12) #14. (Prognosis/Broad[filter]) AND ((thyroidectomy OR thyroidectomies OR strumectomy OR "thyroid surgery" OR "thyroid resection" OR "thyroid removal") AND (osteoporosis OR osteopenia OR "bone density" OR fracture OR fractures)) #15. (#13 OR #14) #16. ((animals [mh:noexp] OR animal experimentation[mesh] OR animal models[mesh])) #17. animals[mesh] AND human[mesh] #18. (#16 NOT #17) #19. (#15 NOT #18) #20. (#13 OR #14) Filters: Humans #21. (#19 OR #20) #22. (#19 OR #20) Filters: English DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 421 UPDATED DATE OF SEARCH: February 24, 2021

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	NUMBER OF ADDITIONAL RESULTS: 20
EMBASE	#1. 'thyroid surgery'/exp #2. thyroidectomy:ti,ab OR thyroidectomies:ti,ab OR strumectomy:ti,ab OR thyroparathyroidectomy:ti,ab #3. 'thyroid gland'/de OR thyroid:ti,ab #4. resection:ti,ab,kw OR surgery:ti,ab,kw OR removal:ti,ab,kw OR operation:ti,ab,kw OR remove:ti,ab,kw #5. #3 AND #4 #6. #1 OR #2 OR #5 #7. 'osteoporosis'/exp #8. osteoporosis:ti,ab OR osteoporoses:ti,ab OR osteoporotic:ti,ab OR 'female athlete triad':ti,ab OR 'bone loss':ti,ab OR 'bone losses':ti,ab #9. 'osteopenia'/exp #10. osteopenia:ti,ab OR osteopenias:ti,ab #11. 'bone density'/exp #12. (bone NEAR/2 density) OR (bone NEAR/2 densities) OR 'mineral densit*' OR 'osseous densit*' OR 'bone mineral content*' OR (bmd:ti,ab AND bone:ti,ab) #13. 'fracture'/exp #14. fracture:ti,ab OR fractures:ti,ab OR 'broken bone*':ti,ab #15. #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 #16. #6 AND #15 #17. ('animal'/exp OR 'invertebrate'/exp OR 'animal experiment'/exp OR 'animal model'/exp OR 'animal tissue'/exp OR 'animal cell'/exp OR 'nonhuman'/exp) NOT ('human'/de OR 'normal human'/de OR 'human cell'/de) #18. #16 NOT #17 #19. #16 AND [humans]/lim #20. #18 OR #19 #21. (#18 OR #19) AND ([english]/lim) DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 1,010 UPDATED DATE OF SEARCH: February 24, 2021 NUMBER OF ADDITIONAL RESULTS: 60

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Cochrane Library	#1. [mh thyroidectomy] #2. (Thyroidectomy or Thyroidectomies or strumectomy):ti,ab #3. [mh thyroid] or thyroid:ti,ab #4. (resection or surgery or removal or operation or remove):ti,ab,kw #5. #3 and #4 #6. #1 or #2 or #5 #7. [mh osteoporosis] #8. (Osteoporosis or Osteoporoses or Osteoporotic or "Female Athlete Triad" or "Bone Loss" or "Bone Losses"):ti,ab #9. (Osteopenia or Osteopenias):ti,ab #10. [mh "Bone Density"] #11. ("Bone density" or "Bone densities" or "mineral density" or "mineral densities" or "osseous density" or "osseous densities" or "bone mineral content" or "bone mineral contents" or (BMD and bone)):ti,ab #12. [mh "bone fractures"] #13. (fracture or fractures or "broken bone" or "broken bones"):ti,ab #14. {or #7-#13} #15. #6 and #14 DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 39 UPDATED DATE OF SEARCH: February 24, 2021 NUMBER OF ADDITIONAL RESULTS: 2
Web Of Science	#1. TS=(thyroidectomy OR thyroidectomies OR strumectomy) #2. TS=(thyroid AND (resection OR surgery OR removal OR operation OR remove)) #3. #1 OR #2 #4. TS=(osteoporosis OR osteoporoses OR osteoporotic OR "female athlete triad" or "bone loss" OR "bone losses") #5. TS=(osteopenia OR osteopenias) #6. TS=("bone density" OR "bone densities" OR "mineral density" OR "mineral densities" OR "osseous density" OR "osseous densities" OR "bone mineral content" OR "bone mineral contents" OR (BMD AND bone))
	#7. TS=(fracture OR fractures OR "broken bone" OR "broken bones")

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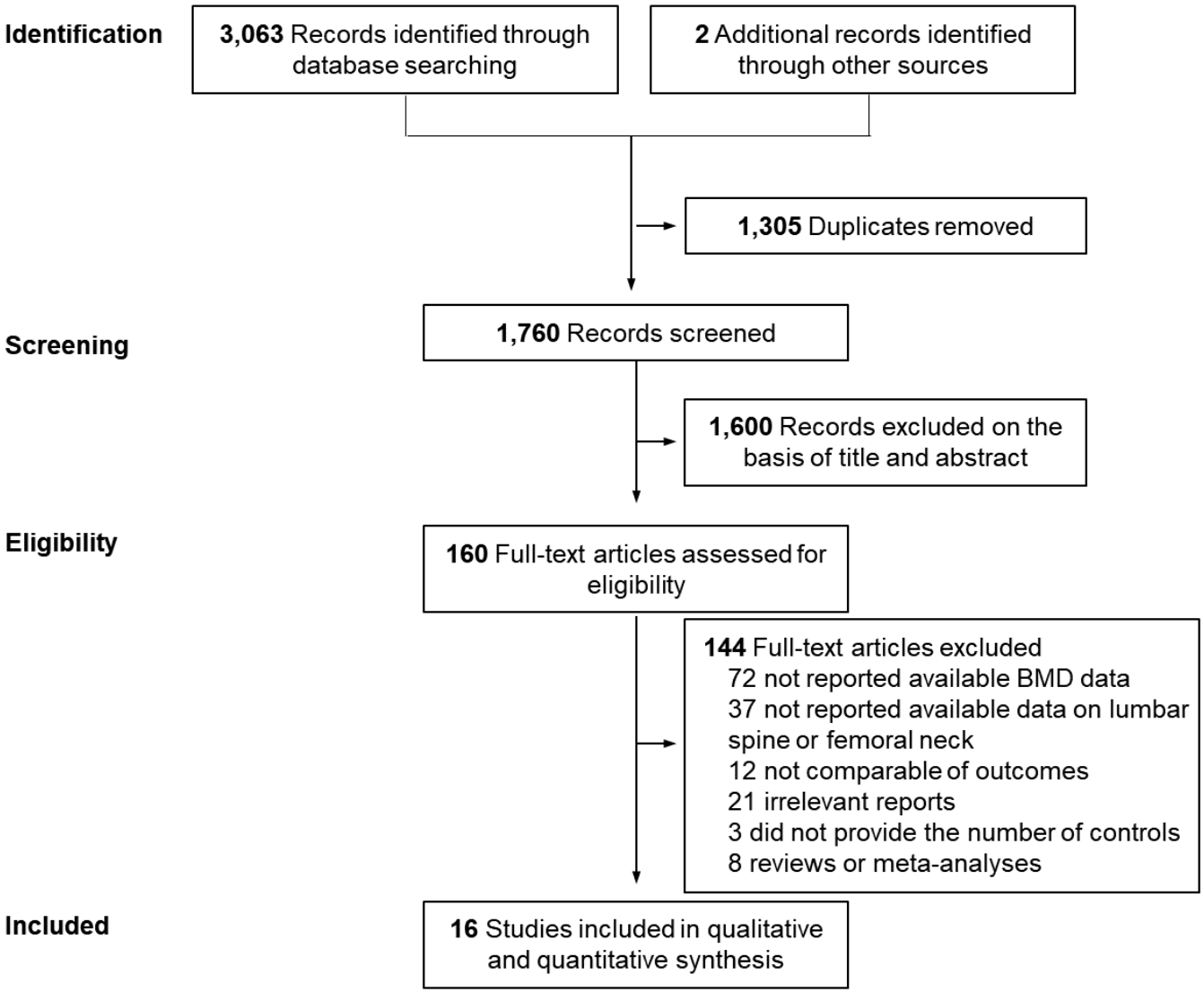
	#8. #4 OR #5 OR #6 OR #7 #9. #3 AND #8 #10. (#3 AND #8) AND LANGUAGE: (English) DATE OF SEARCH: February 24, 2021 NUMBER OF RESULTS: 370
Scopus	#1. TITLE-ABS-KEY(thyroidectomy OR thyroidectomies OR strumectomy) #2. TITLE-ABS-KEY(thyroid AND (resection OR surgery OR removal OR operation OR remove)) #3. #1 OR #2 #4. TITLE-ABS-KEY(osteoporosis OR osteoporoses OR osteoporotic OR "female athlete triad" or "bone loss" OR "bone losses") #5. TITLE-ABS-KEY(osteopenia OR osteopenias) #6. TITLE-ABS-KEY("bone density" OR "bone densities" OR "mineral density" OR "mineral densities" OR "osseous density" OR "osseous densities" OR "bone mineral content" OR "bone mineral contents" OR (BMD AND bone)) #7. TITLE-ABS-KEY(fracture OR fractures OR "broken bone" OR "broken bones") #8. #4 OR #5 OR #6 OR #7 #9. #3 AND #8 #10. (#3 AND #8) AND LANGUAGE(English) DATE OF SEARCH: February 24, 2021 NUMBER OF RESULTS: 1,141
Related article search	Related articles searched according to the following references: 1) Quan ML, Pasieka JL, Rorstad O. Bone mineral density in well-differentiated thyroid cancer patients treated with suppressive thyroxine: a systematic overview of the literature. *J Surg Oncol.* 2002;79(1):62-69; discussion 69-70. 2) Heemstra KA, Hamdy NA, Romijn JA, Smit JW. The effects of thyrotropin-suppressive therapy on bone metabolism in patients with well-differentiated thyroid carcinoma. *Thyroid.* 2006;16(6):583-591. 3) Parker WA, Edafe O, Balasubramanian SP. Long-term treatment-related morbidity in differentiated thyroid cancer: a systematic review of the literature. *Pragmat Obs Res.* 2017;8:57-67. 4) Uzzan B, Campos J, Cucherat M, Nony P, Boissel JP, Perret GY. Effects on bone mass of long term treatment with thyroid hormones: a meta-analysis. *J Clin*

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	Endocrinol Metab. 1996;81(12):4278-4289. 5) Yoon B-H, Lee Y, Oh HJ, Kim SH, Lee Y-K. Influence of thyroid-stimulating hormone suppression therapy on bone mineral density in patients with differentiated thyroid cancer: a meta-analysis DATE OF SEARCH FOR RELATED ARTICLES: April 20, 2020 TOTAL: 332 NUMBER OF RESULTS: 2

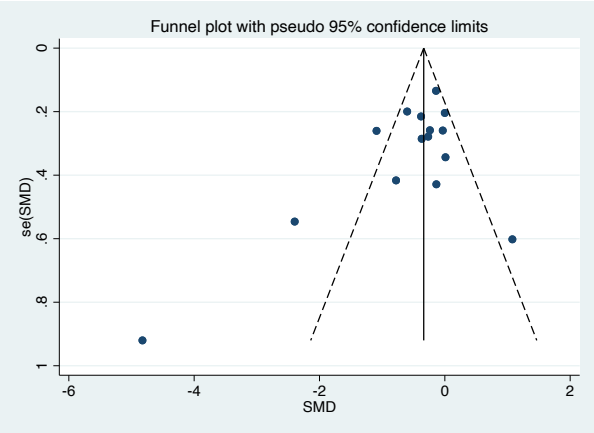
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Supplemental Figure 1. Flow diagram of reference selection process

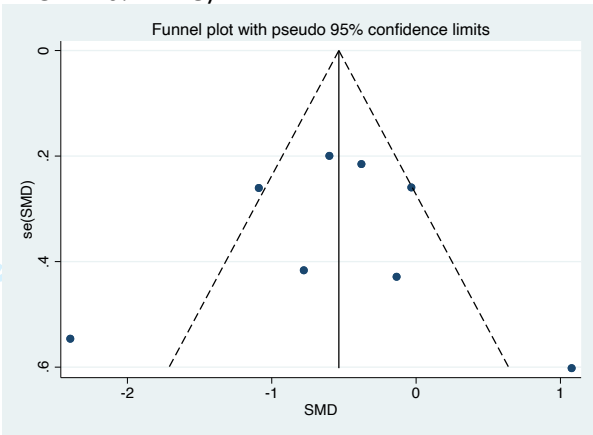


Supplemental Figure 2a. Funnel plots for evaluation of publication bias (Lumbar spine)

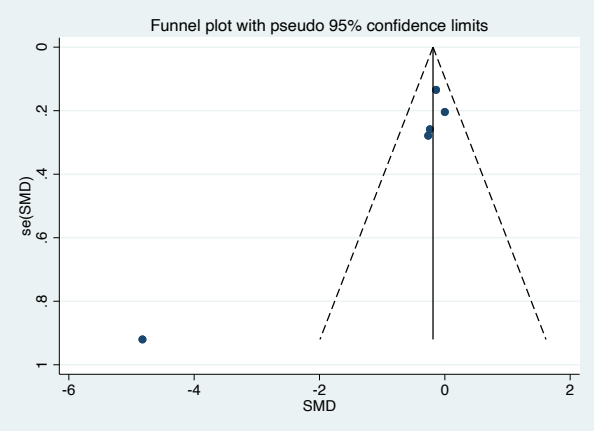
Overall



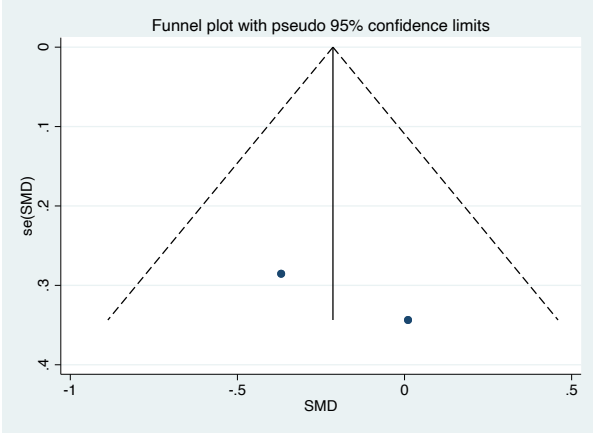
TSH <0.1 mIU/L



TSH 0.1-0.49 mIU/L



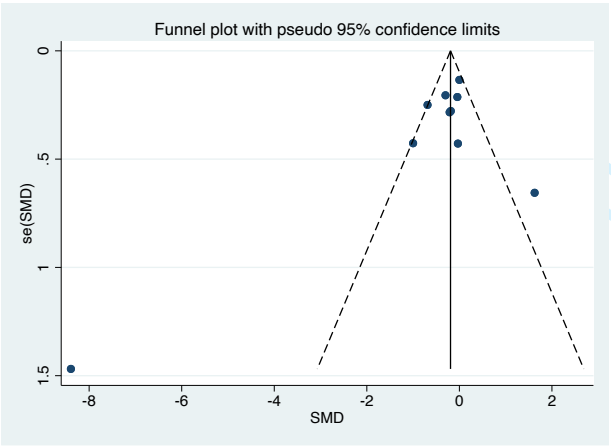
TSH ≥0.50 mIU/L



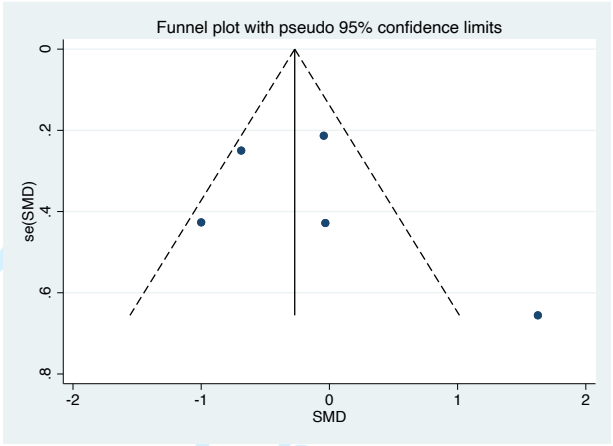
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Supplemental Figure 2b. Funnel plots for evaluation of publication bias (Femoral neck)

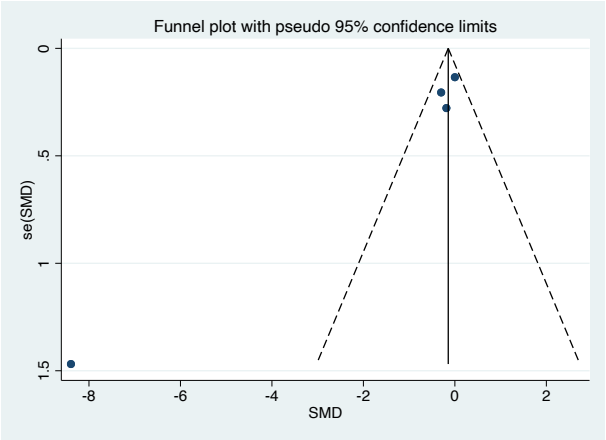
Overall



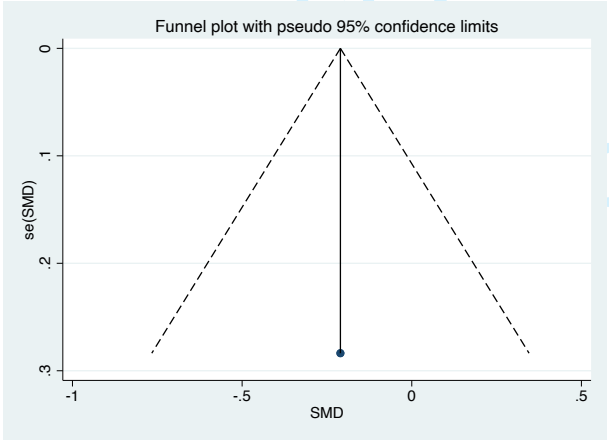
TSH <0.1 mIU/L



TSH 0.1-0.49 mIU/L



TSH ≥0.50 mIU/L



Supplemental Table 2. Additional information on the included studies

First author (year)	Postoperative levothyroxine dose, mean	TSH level at follow-up period, mean	Lists of exclusion criteria for medications*	Lists of exclusion criteria for medical conditions†
Recker (1989)	Range, 0.15-0.20 mg/day	Patients, 1.8 (SD, 0.9) mIU/L; Controls, 3.2 (SD, 1.6) mIU/L	Estrogens, corticosteroids, calcium	Metastatic thyroid cancer, diabetes, post-oophorectomy, post-hysterectomy
Diamond (1991)	Cumulative dose of levothyroxine, mg: 337 (SE, 72)	‡ Patients, 0.3 (SE, 0.1) mIU/L; Controls, 1.2 (SE, 0.2) mIU/L	Estrogens, calcium, vitamin D	(Stated without the list of medical conditions)
Franklyn (1992)	175 (range, 100-200) µg/day	Patients, 0.26 (SD, 0.54) mIU/L; Controls, 1.94 (SD, 1.32) mIU/L	Estrogens, calcium, vitamin D, tamoxifen, thiazide diuretics	Recurrent or metastatic thyroid cancer, osteoporotic fracture, rheumatoid arthritis, diabetes, hypoparathyroidism, alcohol abuse, chronic amenorrhea, late menarche, early menopause, post-oophorectomy
Abugassa (1993)	0.23 (SE, 0.02) mg/day	<0.1 mIU/L	Estrogens	Osteoporotic fracture, endocrinological (apart from hyperparathyroidism), hepatic, intestinal, or renal abnormalities
Kung (1993)	179 (SD, 60) µg/day	Patients, <0.05 mIU/L; Controls, 2.2 (SD, 0.8) mIU/L	Calcium, hormonal contraceptive agents	Recurrent thyroid cancer, late menopause, early menarche, post-oophorectomy, history of other thyroid disorder, family history of osteoporosis
Giannini (1994)	144.2 (SE, 4.15) µg/day	<0.1 mIU/L	(Stated without the list of drugs)	Metastatic thyroid cancer, hypoparathyroidism
Hawkins (1994)	158.3 (SD, 43.7) µg/day	0.3 (SD, 0.4) mIU/L	Estrogens, calcium, vitamin D, thiazide diuretics	(Stated without the list of medical conditions)
Frusciante (1998)	133.6 (SE, 8.4) µg/day	0.04 (SE, 0.016) mIU/L	(Stated without the list of drugs)	(Stated without the list of medical conditions)
Toivonen	215 (SD, 53) µg/day	Patients, <0.05	(Stated without the list of drugs)	(Stated without the list of medical conditions)

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(1998)		mIU/L; Controls, 2.4 (range, 0.06-4.1) mIU/L		
Goerres (1998)	NR ††	0.019 (SD, 0.056) mIU/L	Calcitonin, calcium, vitamin D compounds	NR
Chen (2004)	Starting dose: young patients, 2.0 µg/kg/day; elderly patients, 1.5 µg/kg/day **	1.76 (SD, 0.41) mIU/L	Hormone replacement therapy, corticosteroids, tamoxifen, anticonvulsants, heparin, lithium	Metastatic or recurrent thyroid cancer, hyperparathyroidism, hypoparathyroidism, gonadal insufficiency, diabetes, Cushing's syndrome, severe liver or kidney disorder, chronic obstructive pulmonary disease, rheumatoid arthritis.
Reverter (2005)	NR	Patients, 0.04 (SD, 0.03) mIU/L; Controls, 1.8 (SD, 1.0) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, bisphosphonates, raloxifene, tamoxifen, thiazide diuretics,	Prolonged immobilization, recent bone fracture, inflammatory osteoarticular disease, diabetes, serum creatinine levels >1.3 mg/dL, increased alkaline phosphatase
Eftekhari (2008)	168.33 (SD, 24.3) µg/day	< 0.3 mIU/L	Estrogens, corticosteroids, calcium, vitamin D, bisphosphonates, tamoxifen, thiazide diuretics, anticonvulsants, heparin,	< 45 years, osteoporotic fracture, diabetes, alcohol abuse, rheumatoid arthritis, chronic amenorrhea (> 3 months), late menarche, early menopause, post oophorectomy
Tournis (2015)	122.9 (SD, 21.8) µg/day	Patients, 0.11 (SD, 0.2) mIU/L; Controls, 2.28 (SD, 1.2) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, vitamin D analogues, bisphosphonates, SERMS, denosumab, teriparatide, strontium ranelate, oral contraceptives, diuretics, lithium	Liver or renal disease, prior hyperthyroidism, hyperparathyroidism, hypoparathyroidism, malabsorption syndrome and rheumatic diseases, previous osteoporotic fracture
Moon (2016)	133.5 (SD, 33.5) µg/day	0.09 (SD, 0.13) mIU/L	Bisphosphonate, oral contraceptives, menopausal hormone therapy, SERMS, diuretics, lithium, corticosteroids	Liver or renal diseases, hyperthyroidism, hyper- or hypoparathyroidism, malabsorption syndrome, rheumatic diseases
Kim (2019)	135 (SD, 3.06) µg/day	Patients, 0.000 (IQR 0.00 to 0.242); Controls, 2.450 (IQR 1.263 to 3.813)	Glucocorticoids, anti-osteoporotic drugs	Renal or hepatic impairment

TSH = thyroid stimulating hormone; NR = not reported; NA = not applicable; SD = standard deviation; SE = standard error; SERMS = selective estrogen receptor modulators; hypoPT = patients with hypoparathyroidism; DTC = differentiated thyroid carcinoma; IQR = interquartile range; NTG = non-toxic goiter

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[*] Sixteen studies presented the list of excluded medications, and 6 studies stated that patients who had taken any medication influencing bone density were excluded from the study.

[†] Sixteen studies presented the list of excluded comorbidities, and 6 studies stated that patients who had any medical condition influencing bone density were excluded from the study.

[‡] Lower limit of detection for TSH in this assay was 0.3 mIU/L.

[§] Lower limit of detection for TSH in this assay was 0.15 mIU/L.

[‖] Among 50 participants, 12 were treated with dihydrotachysterol and another 2 with calcitriol, besides calcium tablets. Two patients had well-controlled type 2 diabetes. Three women had been taking hormonal contraceptives for 1 or 3 years, and another had a received low-dose post-menopausal estrogen substitution.

[¶] Lower limit of detection for TSH in this study was 0.2 mIU/L.

[††] Mean cumulated T4 dose of patients was reported as 8527.3 (6102.6) µg/kg body weight.

[**] The dose was adjusted at an increment/decrement of 12.5 to 25 µg/day according to the TSH value at 8- to 12-week intervals.

[‡‡] Levothyroxine dose was adjusted to keep the TSH level under or equal to 0.1 mIU/L.

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Supplemental Table 3. Quality Assessment of Included Studies in the Meta-analysis Using the Newcastle-Ottawa Quality Assessment Scale

First author (year)	Patient selection				Comparability	Exposure			Sum of score
	Adequate case definition	Representativeness of the cases	Selection of controls	Definition of controls		Assessment of Exposure	Same method of ascertainment	Non-Response rate	
Recker (1989)	*	*	*	*	*	-	-	*	6
Diamond (1991)	*	*	*	*	*	-	-	*	6
Franklyn (1992)	*	*	*	*	*	-	-	*	6
Abugassa (1993)	*	*	*	*	**	-	-	*	7
Kung (1993)	*	*	*	*	*	-	-	*	6
Giannini (1994)	*	*	*	*	*	*	-	*	7
Hawkins (1994)	*	*	*	*	*	-	-	*	6
Frusciante (1998)	*	*	*	*	*	*	-	*	7
Toivonen (1998)	*	*	*	*	*	*	-	*	7
Goerres (1998)	*	*	*	*	*	*	-	*	7
Chen (2004)	*	*	*	*	*	*	-	*	7
Reverter (2005)	*	*	*	*	**	*	*	*	9
Eftekhari (2008)	*	*	*	*	**	*	*	*	9
Tournis (2015)	*	*	*	*	**	*	*	*	9
Moon (2016)	*	*	*	*	**	*	*	*	9
Kim (2019)	*	*	*	*	**	*	*	*	9

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	#1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	#2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	#4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	#5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	#6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	#7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	#6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplemental Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Supplemental Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	#7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	#7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	#8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	#8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	#8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	#8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	#8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Supplemental Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	#9-13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	#18-19
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Figure 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Figure 1
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Figure 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	#21
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	#24-25
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	#26
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	#26

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal women: A systematic review and meta-analysis

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Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal women: A systematic review and meta-analysis

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Abstract

Objectives: We assessed thyroid stimulating hormone (TSH) suppression effects on bone mineral density (BMD) in postmenopausal women who underwent thyroidectomy.

Data sources: PubMed, EMBASE, Cochrane Library, Web of Science, and SCOPUS were searched from inception to February 24, 2021.

Study selection: Case control studies were included.

Data extraction and synthesis: Two authors independently reviewed the studies, extracted the data, and performed meta-analysis of eligible studies.

Research design and methods: Studies evaluating BMD in postmenopausal women with thyroid cancer who had thyroidectomy and levothyroxine therapy were included. Differences in BMD were presented as standardized mean differences (SMDs). Meta-analyses were conducted using a random-effects model.

Results: Analysis of 16 case control studies (426 patients and 701 controls without thyroid cancer) showed that stringent TSH suppression (TSH <0.10 mIU/L) after thyroidectomy had deleterious effects on the BMD of the lumbar spine in postmenopausal women compared to controls (SMD, -0.55; 95% confidence interval [CI], -0.99 to -0.10; $I^2 = 75.8\%$). There was no significant difference in patients with moderate TSH suppression (TSH, 0.10–0.49 mIU/L). TSH suppression in postmenopausal women was not significantly associated with lower femoral neck BMD. Subgroup analysis of the lumbar spine showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (SMD, -0.32; 95% CI, -0.50 to -0.14). Subgroup analysis of the femoral neck showed that total thyroidectomy was related to detrimental effects on the BMD of the femoral neck (SMD, -0.60;

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95% CI, -0.89 to -0.31; I2 = 90.4%), but near-total thyroidectomy was not (SMD, 0.00; 95% CI, -0.30 to 0.30; I2 = 55.6%).

Conclusions: Stringent TSH suppression had deleterious effects on the BMD of the lumbar spine after thyroidectomy in postmenopausal women. Further studies are needed to determine whether stringent TSH suppression after thyroidectomy increases the fracture risk.

PROSPERO registration: CRD42018104282

Keywords: thyroid cancer, thyroidectomy, bone mineral density, thyroid stimulating hormone, thyroid stimulating hormone suppression

Strengths and limitations of this study

- This is the first systematic review and meta-analysis to qualitatively collate and assess the results of studies that investigated the effect of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal women.
- We investigated the association between thyroid stimulating hormone suppression and lumbar spine or femoral neck bone mineral density in different subgroups (follow-up duration, surgical extent, ethnicity, and body mass index).
- A possible limitation is that only a few prospective studies were included, and the effects of notable confounding factors and possible fluctuations in the postoperative thyroid stimulating hormone levels were not reflected in the analysis.

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INTRODUCTION

Thyroid stimulating hormone (TSH) levels are suppressed after thyroidectomy for thyroid cancer.[1-3] It is necessary to assess the risk-benefit and adverse effects of TSH suppression because patients with thyroid cancer have a favorable prognosis.[4] To prevent cancer recurrence and reduce cancer-related mortality, levothyroxine is used to achieve TSH suppression,[5] as it inhibits the production of circulating TSH, a factor known to stimulate the proliferation of thyrocytes.[6] However, supraphysiological doses of exogenous levothyroxine may cause side effects, particularly regarding bone health.[2,7] Endogenous hyperthyroidism increases the risk of osteoporosis and osteoporotic fractures.[8,9] However, the effects of TSH suppression due to levothyroxine administration after thyroidectomy are controversial.[10] Some studies reported a lack of deleterious effects on the bone mineral density (BMD) of long-term thyroxine suppressive therapy for differentiated thyroid carcinoma.[11-14] In contrast, other studies reported that thyroxine suppressive therapy had a detrimental effect on BMD.[15,16]

Although different levels and durations of TSH suppression are recommended post-thyroidectomy depending on the risk of thyroid cancer recurrence,[1, 17, 18] the impact of varying levels of TSH suppression on bone health is unclear. Several studies have evaluated the association between TSH suppression levels and fractures.[8, 19, 20] One of them reported that women with a low TSH level (<0.1 mU/L) had a threefold increased risk for hip fractures and a fourfold increased risk for vertebral fractures compared with women who had normal TSH levels.[20] However, this study focused on the risk of fracture rather than BMD in women with low serum TSH levels. Several literature reviews have reported that suppressed TSH levels lead to osteoporosis and decreased BMD after thyroidectomy for thyroid cancer in postmenopausal

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women.[21-26] However, no further studies have investigated the effects of varying TSH levels on BMD.

In this study, we performed a systematic review and meta-analysis of the current literature to investigate the effects of different levels of TSH suppression after thyroidectomy on postoperative BMD in postmenopausal women with thyroid cancer.

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MATERIALS AND METHODS

An expert panel was formed to define the scope of the meta-analysis, literature search, eligibility criteria, and statistical methods according to the preferred reporting items for systematic reviews and meta-analysis guidelines.[27] This systematic review and meta-analysis protocol was registered in the PROSPERO database (registration number, CRD42018104282). At the time of initial registration of PROSPERO, patients who underwent thyroidectomy due to any thyroid disease were registered. However, there are few studies on BMD after thyroidectomy in typical hyperthyroidism diseases such as Graves' disease, toxic multinodular goiter, and toxic adenoma, excluding cancer, so our study was limited to only patients with thyroid cancer.

Patient and public involvement statement

This meta-analysis was based on study-level data, and no individual-level data were involved in the study or in defining the research question or outcome measures.

Ethics approval statement

This study was approved by the Institutional Review Board of Korea University Anam Hospital (registration number: 2021AN0165).

Literature search

A trained librarian searched the PubMed, EMBASE, Cochrane Library, Web of Science, and SCOPUS databases to identify relevant studies published before February 24, 2021 (Supplemental Table 1). The following search terms, adapted to each database, were used: "*thyroidectomy*," "*levothyroxine*," "*thyroid stimulating hormone*," "*bone density*,"

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"*osteoporosis*," and "*bone fractures*." We also included studies reported in earlier systematic reviews,[21-24, 26] and those identified from the bibliography of the included articles.

Eligibility criteria and study selection

The inclusion criteria were as follows: (1) all participants underwent thyroidectomy for thyroid cancer, followed by the administration of levothyroxine; (2) availability of BMD data, irrespective of whether this was the primary endpoint; (3) availability of data on the lumbar spine and femoral neck; (4) comparison of postoperative outcomes between the study and control groups (matched healthy postmenopausal women were enrolled); and (5) the most recent or informative publication data, in the case of multiple publications from the same population. The exclusion criteria were as follows: (1) only abstract available; (2) case reports, reviews, and editorials; (3) studies with pediatric or pregnant populations; (4) publication language other than English; and (5) missing measures of variability (standard deviations, standard errors, or interquartile ranges) when effect estimates were presented.

Two authors (D.K. and Y.K.) independently assessed the eligibility of all studies identified through an electronic literature search based on the titles and abstracts. Eligible articles with full texts were further evaluated for inclusion. Disagreements were resolved by consensus.

Data extraction and risk-of-bias assessment

A standard template suggested by Cochrane was adapted and applied for data extraction.[28] Two authors (D.K. and J.H.) independently extracted the data, including the first author's name; year of publication; study location; study design; age, body mass index (BMI), ethnicity, number of the study participants; etiology; extent of thyroidectomy; location and value

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of the BMD measurement; follow-up duration after thyroidectomy; levothyroxine dose; preoperative and postoperative laboratory results including TSH, free thyroxine, parathyroid hormone, and calcium levels; medications used by the study participants; and comorbidities.

The quality of the included studies was assessed using the Newcastle-Ottawa Quality Assessment Scale.[29] This assesses the quality of nonrandomized studies according to three variables: selection of the study population, comparability, and exposure factor.

Data synthesis and analysis

Each outcome was presented with estimates calculated as standardized mean differences (SMDs). Assuming that the baseline values of patients and controls without thyroid cancer were balanced and the confounders were controlled by the study design, we compared the measurements at the end of the follow-up period. The SMDs were coded so that positive values represented a higher BMD in the thyroidectomy group than in the control group, with 0.2, 0.5, and 0.8, corresponding to small, moderate, and large effects, respectively.[30] We estimated the pooled SMDs and 95% confidence intervals (CIs) using the random-effects model (DerSimonian and Laird method). Statistical significance was set at a two-tailed p-value of <0.05. We assessed heterogeneity using the Mantel–Haenszel model and I^2 values (the percentage of variance in the pooled estimate due to between-study differences), with I^2 thresholds of ≤25% (low), 26%–50% (moderate), and >50% (high).

Subgroup analyses were performed considering the TSH level (categorized as stringent suppression, <0.10 mIU/L; moderate suppression, 0.10–0.49 mIU/L; and lower normal, 0.50–2.0 mIU/L). Other investigated factors included the follow-up duration, baseline age, baseline BMI, ethnicity, and extent of thyroidectomy (total vs. near-total). Publication bias was assessed based

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on funnel plot asymmetry using Begg's and Egger's tests at a p-value of < 0.1. Stata version 13.1 software (StataCorp., College Station, TX, USA) was used for all statistical analyses.

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RESULTS

Studies included in the meta-analysis

A systematic literature search retrieved a total of 3,063 studies (Supplemental Figure 1). We included two studies from the bibliography of the included articles. Of the included studies, 1,305 duplicate studies were excluded. The remaining 1,760 articles were screened based on the title and abstract and assessed for eligibility. The full text of 169 potentially eligible studies was evaluated by two independent authors (D.K. and J.H.), of which 153 studies did not meet the inclusion criteria (reasons for exclusion are summarized in Supplemental Figure 1). Sixteen studies,[11-16, 31-40] including 1,127 postmenopausal women, met the inclusion criteria. Table 1 lists the summary and characteristics of each study.

TABLE 1. Characteristics of the studies included in the analysis

First author (year, location)	Study population				Methods		Outcomes
	Age, mean (SD), years	Etiology	Extent of thyroidectomy	No. of postmenopausal participants, n (total=1,127)	Assessment tool of bone density (measurement site included in the meta-analysis)	Outcomes included in the meta-analysis (unit)	Assessment time after thyroidectomy, mean, year
Recker (1989, USA)[31]	50.2	DTC	Total	Patients, 18; Controls, 16	DPA (lumbar spine and femoral regions)	BMD (g/m^2)	9.6

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Diamond (1991, Australia) [16]	59 (2.8)	DTC	Total	Patients, 10; Controls, 10	DPA (Lumbar spine, femoral neck)	BMD (g/m^2)	5.9
Franklyn (1992, UK) [32]	63.4 (8.1)	DTC	Subtotal[d]	Patients, 26; Controls, 26	DEXA (lumbar spine and femoral regions)	BMD (g/m^2)	NR
Abugassa (1993, Sweden) [33]	58 (2)	Thyroid cancer	Total	Patients, 13; Controls, 12	DPA (lumbar spine and femoral regions)	BMD (g/cm^2)	13.0
Kung (1993, China) [15]	62 (8)	Thyroid cancer	Total	Patient, 34; control, 34	DEXA (lumbar spine, femoral regions)	BMD (g/cm^2)	12.2
Giannini (1994, Italy) [34\]	Patients, 57.6 (1.7); Control, 56 (0.4)	Non-medullary thyroid carcinoma[c]	Total	Patients, 13; Controls, 11	DEXA (lumbar spine, femoral regions)	BMD (g/cm^2)	7.6

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Hawkins (1994, Spain) [35]	Patients, 59.6 (7.5); Controls, 49.3 (5.2)	Thyroid cancer	Near-total	Patients, 21; Controls, 53	DEXA (lumbar spine)	BMD (g/cm^2)	6.2
Frusciante (1998, Italy) [11]	Patients, 55.3 (1.7); Controls, 53.3 (2.5)	DTC	Near-total	Patients, 6; Controls, 7	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.7
Toivonen (1998, Finland) [12]	Patients, 45 (range, 27 to 71); Controls, 43 (range, 26 to 65) [a]	DTC	Near-total	Patients, 10; Controls, 12	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	10
Goerres (1998, Switzerland) [14]	Patients, 61.6 (12.0); Control	Thyroid cancer	Total	Patients, 23; Controls, 42	DEXA (lumbar spine,	BMD (g/cm^2)	8.8

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	s, 562.7 (8.1)				femoral neck)		
Chen (2004, China) [36]	57.7 (6.9)	DTC	Near-total	Patients, 25; Controls, 25	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	7.8
Reverter (2005, Spain) [13]	58 (9)	DTC	Near-total	Patients, 44; Controls, 44	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	12.0
Eftekhari (2008, Iran) [37]	Patients, 51.7 (7.3); Controls, 51.2 (7.8) [a]	DTC	NR	Patients, 33; Controls, 33	DEXA (lumbar spine)	BMD (g/cm^2)	1.2
Tournis (2015, Greece) [38]	Patients, 56.7 (5.9); Controls, 57.8 (6.1)	DTC	Total	Patients, 40; Controls, 60	DEXA (lumbar spine, femoral neck)	BMD (mg/cm^3) [b]	5.0

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Moon (2016, Korea) [39]	Patients, 61.4 (7.6); Controls, 61.4 (7.6)	DTC	NR	Patients, 74; Controls, 222	DEXA (lumbar spine, femoral neck)	BMD (g/cm^2)	5.2
Kim (2019, Korea) [40]	Patients, 53; Controls, 55	DTC	Total	Patients, 36; Controls, 94	DEXA (lumbar spine)	BMD (g/cm^2)	4.9

SD = standard deviation; NR = not reported; DPA = dual photon absorptiometry; DEXA = dual-energy x-ray absorptiometry; DTC = differentiated thyroid carcinoma; BMD = bone mineral density

[a] These studies presented the mean or median age of the total study population, including pre- and postmenopausal women.

[b] This value was calculated with the method of volumetric BMD. For more information, refer to Tournis et al. (2015).

[c] This study involved patients with non-medullary thyroid carcinoma without any kind of drug treatment known to influence bone turnover or diseases affecting calcium metabolism.

[d] This study which involved subjects that underwent subtotal thyroidectomy was excluded from the total versus near-total subgroup analysis.

Studies were published as early as 1989, and the follow-up duration after thyroidectomy ranged from one to 15 years. Three studies were prospective,[13, 16, 38] while three involved consecutive sampling (Table 2).[14, 33, 37]

TABLE 2. Quality rating for studies included in the analysis

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First author (Year)	Design	Sampling strategy	Multisite	Sample size	Method of controlling for confounders	Confounders included in the analysis (Confounders included in the exclusion criteria are presented in the right two columns)	Exclusion criteria for medication [a]	Exclusion criteria for medical conditions [b]
Recker (1989) [31]	Retrospective cohort with matched control	Nonconsecutive	No	34	Matching, Exclusion	NR	Stated with the list of drugs	Stated with the list of medical conditions
Diamond (1991) [16]	Prospective cohort with matched control	Nonconsecutive	No	48	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated
Frankly n	Retrospective cohort with	Nonconsecutive	No	98	Matching, Exclusion	Matching: age, sex, BMI,	Stated with the	Stated with the list of

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(1992) [32]	matched control					menopausal status, smoking history, calcium intake score	list of drugs	medical conditions
Abugassa (1993) [33]	Retrospective cohort with matched control	Consecutive	No	25	Matching, Exclusion	Matching: age, sex, menopausal status, postoperative time, postmenopausal age	Stated with the list of drugs	Stated with the list of medical conditions
Kung (1993) [15]	Retrospective cohort with matched control	Nonconsecutive	No	68	Matching, Exclusion	NR	Stated with the list of drugs	Stated with the list of medical conditions
Giannini (1994) [34]	Retrospective cohort with matched control	Nonconsecutive	No	46	Matching, Exclusion	Matching: sex, height, weight, menopausal status	Stated	Stated

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Hawkins (1994) [35]	Retrospective cohort with matched control	Nonconsecutive	No	74	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated
Fruscia nte (1998) [11]	Retrospective cohort with matched control	Nonconsecutive	No	39	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated	Stated
Toivonen (1998) [12]	Retrospective cohort with matched control	Nonconsecutive	No	67	Matching, Exclusion	Matching: age, sex, menopausal status	Stated	Stated
Goerres (1998) [14]	Retrospective cohort with matched control	Consecutive	No	114	Matching, Exclusion	Matching: age, sex	Stated with the list of drugs	NR
Chen (2004) [36]	Retrospective cohort with matched control	Nonconsecutive	No	50	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions

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Reverter (2005) [13]	Prospective cohort with matched control	Nonconsecutive	No	88	Matching, Exclusion	Matching: age, sex, BMI, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions
Eftekhari (2008) [37]	Retrospective cohort with matched control	Consecutive	No	132	Matching, Exclusion	Matching: age, sex, menopausal status	Stated with the list of drugs	Stated with the list of medical conditions
Tournis (2015) [38]	Prospective cohort with matched control	Nonconsecutive	No	169	Matching, Exclusion	Matching: age, sex, BMI, menopausal status, calcium intake, years since menopause in case of postmenopausal women	Stated with the list of drugs	Stated with the list of medical conditions

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Moon (2016) [39]	Retrospective cohort with matched control	Nonconsecutive	No	496	Matching, Exclusion	Matching: age, sex	Stated with the list of drugs	Stated with the list of medical conditions
Kim (2019) [40]	Retrospective cohort with matched control	Nonconsecutive	No	130	Matching	Matching; age, BMI	NR	NR

NR = not reported; NA = not applicable; BMI = body mass index; TSH = thyroid stimulating hormone; BMD = bone mineral density

[a] Categories include NR (not reported), stated (without the list of drugs), and stated with the list of drugs. The lists of drugs for exclusion are summarized in Supplemental Table 2.

[b] Categories include NR (not reported), stated (without the list of drugs), and stated with the list of drugs. The lists of medical conditions for exclusion are summarized in Supplemental Table 2.

[c] Additional analyses used multivariable regression to evaluate the association between the TSH level and bone density after thyroidectomy controlled for TSH, free thyroxine, urinary calcium, serum alkaline phosphatase, serum osteocalcin, serum telopeptide, BMD of the lumbar spine, and BMD of the femoral hip. The comparison of bone density between patients and controls was performed using a paired *t*-test without controlling for confounders.

All studies, except one,[14] controlled for confounders in the study design using matching factors and exclusion criteria (Table 2 and Supplemental Table 2). The quality scores are shown in Supplemental Table 3, with scores ranging from six to nine. All studies were found to be of

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moderate to high quality. After thyroidectomy, the mean TSH levels ranged between 0.00 and 1.8 mIU/L. The postoperative levothyroxine doses are summarized in Supplemental Table 2.

Effects of TSH suppression on the lumbar spine

All the included studies reported BMD of the lumbar spine in 426 postmenopausal women that underwent thyroidectomy for thyroid cancer and 701 controls. Overall, TSH suppression after thyroidectomy showed detrimental effects on the BMD of the lumbar spine (SMD, -0.46; 95% CI, -0.77 to -0.16; I^2 = 77.6%; p =.003). When analyzed according to the TSH suppression target, stringent suppression, but not moderate suppression, showed negative effects on the postoperative BMD (SMD, -0.55; 95% CI, -0.99 to -0.10; I^2 = 75.8%; p =.016) (Figure 1A). Funnel plots indicated that there was no significant publication bias with stringent TSH suppression (TSH < 0.1 mIU/L) (Supplemental Figure 2A), which was supported by Egger's test (p =0.940).

Effects of TSH suppression on the femoral neck

Ten of the studies (282 postmenopausal women and 452 controls) investigated the effects of TSH suppression after thyroidectomy for thyroid cancer on the BMD of the femoral neck.[11-13, 15, 16, 32, 33, 36, 38, 39] The BMD of the femoral neck was similar in TSH-suppressed postmenopausal women that underwent thyroidectomy for thyroid cancer compared to that in controls (SMD, -0.32; 95% CI, -0.74 to 0.10; I^2 = 81.8%; p =.133; Figure 1B). This observation was similar across subgroups according to the suppressed TSH levels. Funnel plots indicated no significant publication bias (Supplemental Figure 2B), as supported by Egger's test (p =0.177).

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Subgroup analyses

A subgroup analysis showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (Figure 2A). No significant differences in subgroups with different ethnicities and BMIs were detected. Furthermore, the detrimental effects of TSH suppression were more pronounced in patients who underwent total thyroidectomy than in those that underwent near-total thyroidectomy without statistical significance ($p = 0.065$). A significant interaction between surgical extent and BMD of the femoral neck (Figure 2B; p for interaction, 0.004) was observed. Moreover, total thyroidectomy was related to detrimental effects on the BMD of the femoral neck (SMD, -0.60; 95% CI, -0.89 to -0.31; $I^2 = 90.4\%$; $p < 0.001$), but near-total thyroidectomy was not (SMD, 0.00; 95% CI, -0.30 to 0.30; $I^2 = 55.6\%$; $p = 0.996$). The follow-up duration, ethnicity, and BMI did not affect the association between TSH suppression and femoral neck BMD in postmenopausal patients who underwent thyroidectomy for thyroid cancer.

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DISCUSSION

Summary of the findings

The potential effect of hyperthyroidism on postoperative bone health in patients who underwent thyroidectomy for thyroid cancer remains a concern in terms of long-term care. Identifying the groups that are at risk of deteriorating bone health after thyroidectomy is crucial for tailoring TSH suppression and guiding clinical treatment in a more focused and individualized direction. In this meta-analysis of 16 studies, including 1,127 participants, we identified stringent TSH suppression after thyroidectomy for thyroid cancer that adversely affects the postoperative BMD in postmenopausal women. BMD is widely used for osteoporosis screening, and BMD changes are an independent risk factor for fracture.[41] Our findings suggest that the oncological benefits of stringent TSH suppression in postmenopausal women should be weighed against its possible adverse effects on postoperative bone health.

Comparison with other studies

Clinical guidelines recommend the administration of levothyroxine as the optimal treatment for patients undergoing thyroidectomy for thyroid cancer. Patients with residual carcinoma or those at a high risk of cancer recurrence, those at an intermediate risk of cancer recurrence, and those at a low risk of cancer recurrence should maintain TSH levels at <0.1, 0.1–0.5, and 0.5–2 mIU/L, respectively.[1, 17, 18] Due to its beneficial effects on cancer recurrence,[42] and overall survival,[43] stringent TSH suppression is recommended for patients at a high risk of cancer recurrence. However, there is no consensus regarding serum TSH levels below the lower limit of the normal range.[44] Because of the favorable prognosis associated with TSH suppression, patients with thyroid cancer receive life-long thyroxine treatment and have suppressed TSH

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levels for extended periods. Some studies have reported that long-term thyroxine therapy leads to an increased risk of bone fractures.[45, 46] However, the effects of TSH suppression on the bone health of patients undergoing thyroidectomy for thyroid cancer are rarely discussed in the current guidelines, and few studies evaluate the adverse effects according to stratified TSH levels. Several recent population-based studies have reported associations between levothyroxine doses and bone health.[19, 47] One study concluded that among patients with thyroid cancer, those who received levothyroxine had an increased risk of osteoporosis in a cumulative dose-dependent manner compared to those who did not receive levothyroxine (hazard ratio [HR] at a cumulative levothyroxine dose >395 mg, 3.62; 95% CI, 2.16 to 6.06; HR at a cumulative levothyroxine dose ≤265 mg, 1.53; 95% CI, 0.91 to 2.57).[47] In another study, 185,956 patients with thyroid cancer that were administered a levothyroxine dosage of ≥170 μg/day had a higher risk of osteoporotic fractures (HR, 1.25; 95% CI, 1.07 to 1.45) than matched controls without cancer.[19] However, as the intensity of TSH suppression therapy was stratified based on prescribed levothyroxine doses, results from these studies reflect the indirect effects of TSH concentration on the bone.

According to a series of previous studies, TSH suppression after thyroidectomy significantly reduced bone mass in postmenopausal women, whereas it did not affect premenopausal women and men.[21, 24, 48] Menopause and the subsequent decrease in estrogen are well-known risk factors that contribute to osteoporosis.[49] The prominent effects of TSH suppression on bone health in postmenopausal women are clinically significant, and affected subjects need to be appropriately managed. We observed deleterious effects of stringent TSH suppression on the BMD of the lumbar spine. However, moderate TSH suppression did not significantly affect postoperative BMD. Inhibition of the receptor activator of nuclear factor-κB

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ligand-induced osteoclast formation,[50] and activation of Wnt-5a- and β-arrestin 1-mediated osteoblast differentiation support that lower TSH levels may be responsible for lowering BMD.[48, 49]

BMD comparison on the lumbar spine and femoral neck

The detrimental effects of TSH suppression have been reported to be more prominent on the BMD of the lumbar spine than on that of the femoral neck (Figure 1). This result is consistent with that of previous studies. First, the lumbar spine is predominantly a trabecular bone. Its microarchitecture is more vulnerable to osteoporotic changes than that of the cortical bone, which comprises a large portion of the femoral neck.[51] Second, the mineral contents of the lumbar spine change primarily during menopause, while those of the femoral neck change at a constant rate over one's lifetime.[52] These factors may affect the interaction between the BMD of the lumbar spine and femoral neck and TSH and estrogen levels in postmenopausal women. Low BMD of the lumbar spine is clinically significant as the spine is the most common osteoporotic fracture site,[53] and spinal fractures are related to increased mortality.[54] A recent population-based study assessing the risk of fracture after thyroidectomy by tracing patients for 40 years also reported a significant increase in the risk of vertebral fractures (standardized incidence ratio, 2.8; 95% CI, 2.3 to 3.3) and a relatively small increase in the risk of proximal femur fractures (standardized incidence ratio, 1.3; 95% CI, 1.0 to 1.8) after thyroidectomy.[55]

The deleterious effects of TSH suppression on the BMD of the lumbar spine were more prominent in patients who underwent total (vs. near-total) thyroidectomy, in Asian (vs. non-Asian) populations, and in those with a BMI <25 kg/m^2 (vs. BMI ≥25 kg/m^2). Near-total

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thyroidectomy, which leaves a small amount of thyroid tissue (<1.0 mL), can be performed instead of total thyroidectomy, whereas partial or subtotal thyroidectomy is not appropriate for thyroid malignancy.[1] Limited information is available regarding the differential effects of total and near-total thyroidectomy on bone health and BMD, suggesting the need for further research regarding this aspect. A study found that patients with a lower BMI that underwent thyroidectomy because of benign pathology required a significantly higher levothyroxine dose per kilogram to achieve a euthyroid state compared to those with a higher BMI (BMI <25 kg/m^2, 1.84 ± 0.07 μg/kg/day; BMI ≥35 kg/m^2, 1.39 ± 0.05 μg/kg/day); this association was age independent.[56] We suggest levothyroxine dose adjustment and thorough monitoring of the BMD in nonobese patients undergoing thyroidectomy for thyroid cancer, followed by TSH suppression.

Limitations

Our study has several limitations. First, most of the included studies had a retrospective design, and the number of controlled confounding factors was heterogeneous in each study. Furthermore, studies may have been misclassified because of various factors known to affect TSH levels, such as factors related to the postmenopausal state,[57] and parathyroid hormone levels.[58] Second, most of the included studies that involved TSH suppression reported the TSH level at a single time point. Although each study periodically assessed TSH levels to confirm maintenance within the intended level, fluctuations in TSH levels cannot be ruled out, and consequently, the effects of these fluctuations on BMD may represent potential confounders. Third, few studies have considered genetic bone susceptibility or the effects of lifestyle factors such as physical activity, smoking, and alcohol consumption on the BMD. Fourth, some studies

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may have been misclassified because of the use of a first-generation TSH assay with insufficient sensitivity.[59] Fifth, most of the cases were differentiated thyroid carcinoma, but thyroid cancer etiology was not reported, or patients with other etiologies were included. Therefore, additional treatments or differences in molecular pathogenesis may be the cause of heterogeneity. Sixth, most studies did not clarify whether all of the included patients were high risk or suppressed below the TSH level of 0.1 mIU/L. Therefore, statistical analyses were performed based on the average TSH levels presented in each study. Seventh, other variables that may influence hypoparathyroidism or vitamin D levels, such as autoimmune conditions, genetic variants, radiation, mineral deposition, and magnesium deficiency, were not evaluated.[60] Eighth, only case control studies were enrolled. A prospective cohort and longitudinal study are required. Ninth, funnel plots of the lumbar spine indicated that there was some publication bias with overall and moderate TSH suppression (TSH, 0.10–0.49 mIU/L) (Supplemental Figure 2a.), as shown by the Egger test (p =0.098 and 0.082, respectively). Interpretation of the overall and moderate TSH suppression groups requires further investigation. Tenth, few studies have analyzed the osteoporosis and fracture risk results according to the TSH suppression level, so this was not covered in our systematic review. Further investigation is needed regarding the benefits of TSH suppression therapy in relation to cancer recurrence or progression and the risk of osteoporosis or fractures, particularly in patients at a high risk of cancer recurrence receiving stringent TSH suppression therapy.

CONCLUSIONS

Stringent TSH suppression (TSH level, <0.10 mIU/L) after thyroidectomy for thyroid cancer was found to exert detrimental effects on the postoperative BMD of the lumbar spine in

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postmenopausal women. Further well-designed prospective studies are needed to evaluate the change in the BMD of different bones and the risk of bone fractures according to the TSH suppression levels in postmenopausal women who have undergone thyroidectomy for thyroid cancer to establish adequate postoperative management.

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DATA SHARING STATEMENT: Data are available upon request. The data are available upon request from the corresponding author. (kukwon@korea.ac.kr)

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FIGURE LEGENDS

Figure 1. Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women undergoing TSH suppression after thyroidectomy for thyroid cancer.

Meta-analysis results of case-control studies included in this review were presented according to TSH level. SMDs were calculated such that positive values indicated benefit for BMD in thyroidectomized group compared to controls without thyroid cancer.

TSH, thyroid-stimulating hormone; SMD, standardized mean difference; BMD, bone mineral density.

Figure 2. Subgroup analysis

Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women after thyroidectomy for thyroid cancer and subsequent TSH suppression by subgroups.

ES, effect size; BMD, bone mineral density; BMI, body mass index.

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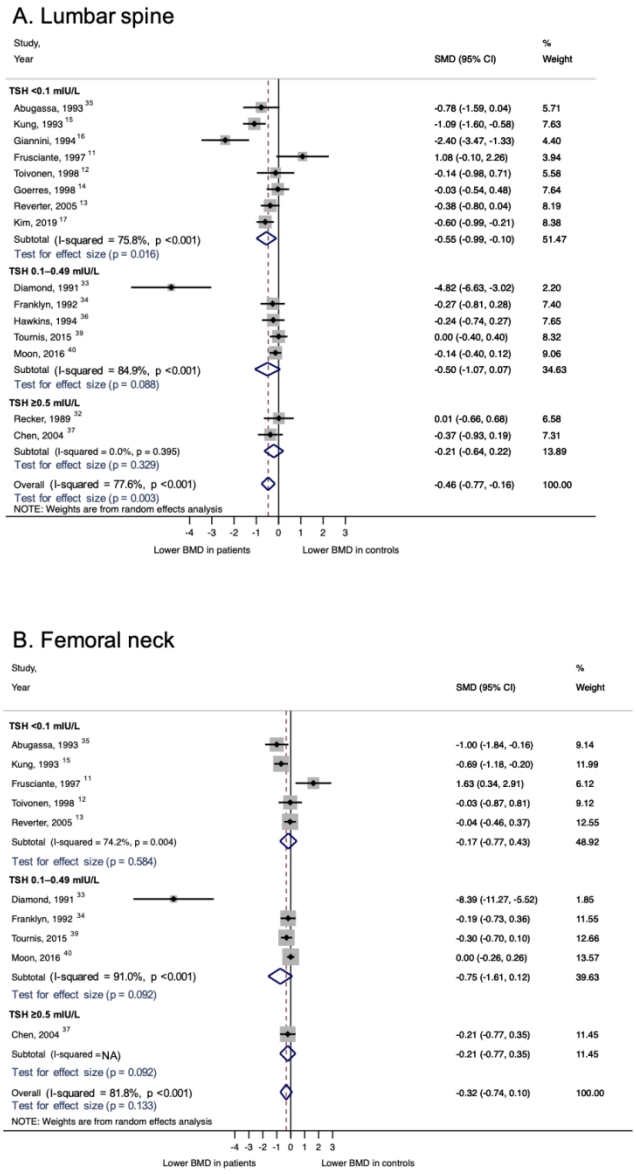


Figure 1. Postoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women undergoing TSH suppression after thyroidectomy for thyroid cancer. Meta-analysis results of case-control studies included in this review were presented according to TSH level. SMDs were calculated such that positive values indicated benefit for BMD in thyroidectomized group compared to controls without thyroid cancer.TSH, thyroid-stimulating hormone; SMD, standardized mean difference; BMD, bone mineral density

345x611mm (72 x 72 DPI)

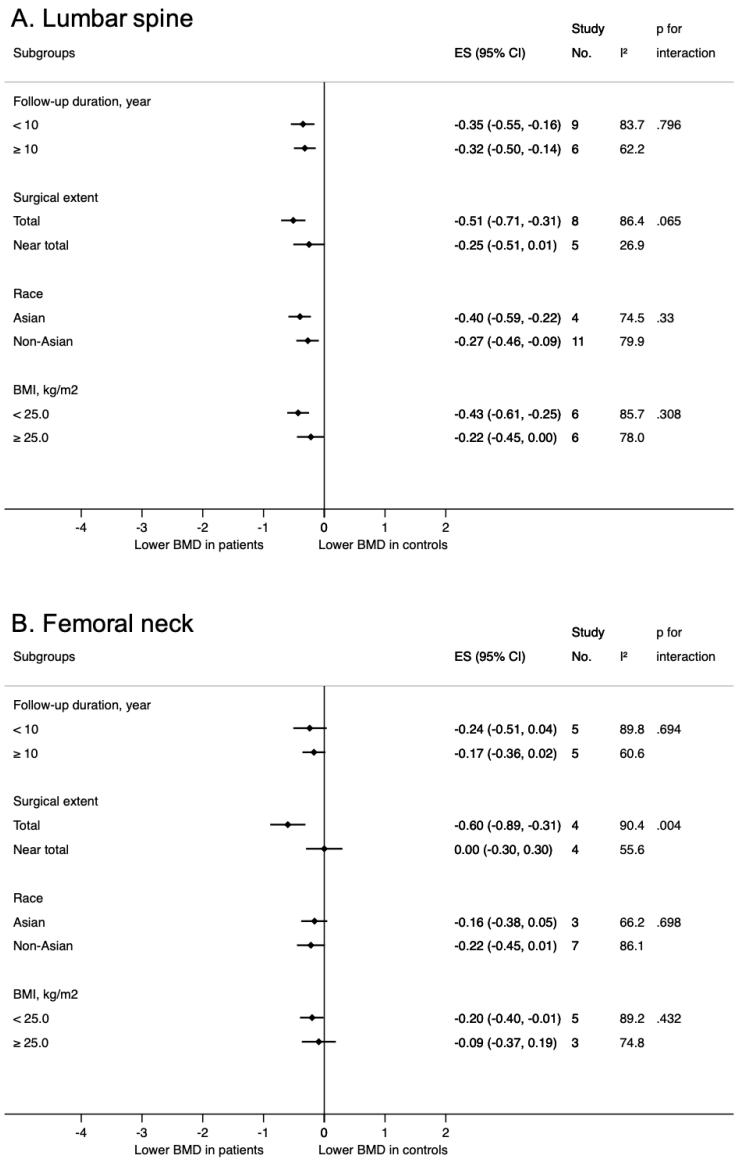


Figure 2. ubgroup analysisPostoperative BMD of lumbar spine (A) and femoral neck (B) in postmenopausal women after thyroidectomy for thyroid cancer and subsequent TSH suppression by subgroups.ES, effect size; BMD, bone mineral density; BMI, body mass index

352x541mm (72 x 72 DPI)

Supplemental Table 1. Search strategy

Database	Search strategy
PubMed	#1. thyroidectomy[mesh] #2. Thyroidectomy[tiab] OR Thyroidectomies[tiab] OR strumectomy[tiab] #3. (thyroid[mesh] OR thyroid[tiab]) AND (resection[tiab] OR surgery[tw] OR removal[tiab] OR operation[tiab] OR remove[tiab]) #4. (#1 OR #2 OR #3) #5. osteoporosis[mesh] #6. Osteoporosis[tiab] OR Osteoporoses[tiab] OR Osteoporotic[tiab] OR "Female Athlete Triad"[tiab] OR "Bone Loss"[tiab] OR "Bone Losses"[tiab] #7. (Osteopenia[tiab] OR Osteopenias[tiab]) #8. Bone Density[mesh] #9. "Bone density"[tiab] OR "Bone densities"[tiab] OR "mineral density"[tiab] OR "mineral densities"[tiab] OR "osseous density"[tiab] OR "osseous densities"[tiab] OR "bone mineral content"[tiab] OR "bone mineral contents"[tiab] OR (BMD[tiab] AND bone[tiab]) #10. bone fractures[mesh] #11. fracture[tiab] OR fractures[tiab] OR "broken bone"[tiab] OR "broken bones"[tiab] #12. (#5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11) #13. (#4 AND #12) #14. (Prognosis/Broad[filter]) AND ((thyroidectomy OR thyroidectomies OR strumectomy OR "thyroid surgery" OR "thyroid resection" OR "thyroid removal") AND (osteoporosis OR osteopenia OR "bone density" OR fracture OR fractures)) #15. (#13 OR #14) #16. ((animals [mh:noexp] OR animal experimentation[mesh] OR animal models[mesh])) #17. animals[mesh] AND human[mesh] #18. (#16 NOT #17) #19. (#15 NOT #18) #20. (#13 OR #14) Filters: Humans #21. (#19 OR #20) #22. (#19 OR #20) Filters: English DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 421 UPDATED DATE OF SEARCH: February 24, 2021

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	NUMBER OF ADDITIONAL RESULTS: 20
EMBASE	#1. 'thyroid surgery'/exp #2. thyroidectomy:ti,ab OR thyroidectomies:ti,ab OR strumectomy:ti,ab OR thyroparathyroidectomy:ti,ab #3. 'thyroid gland'/de OR thyroid:ti,ab #4. resection:ti,ab,kw OR surgery:ti,ab,kw OR removal:ti,ab,kw OR operation:ti,ab,kw OR remove:ti,ab,kw #5. #3 AND #4 #6. #1 OR #2 OR #5 #7. 'osteoporosis'/exp #8. osteoporosis:ti,ab OR osteoporoses:ti,ab OR osteoporotic:ti,ab OR 'female athlete triad':ti,ab OR 'bone loss':ti,ab OR 'bone losses':ti,ab #9. 'osteopenia'/exp #10. osteopenia:ti,ab OR osteopenias:ti,ab #11. 'bone density'/exp #12. (bone NEAR/2 density) OR (bone NEAR/2 densities) OR 'mineral densit*' OR 'osseous densit*' OR 'bone mineral content*' OR (bmd:ti,ab AND bone:ti,ab) #13. 'fracture'/exp #14. fracture:ti,ab OR fractures:ti,ab OR 'broken bone*':ti,ab #15. #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 #16. #6 AND #15 #17. ('animal'/exp OR 'invertebrate'/exp OR 'animal experiment'/exp OR 'animal model'/exp OR 'animal tissue'/exp OR 'animal cell'/exp OR 'nonhuman'/exp) NOT ('human'/de OR 'normal human'/de OR 'human cell'/de) #18. #16 NOT #17 #19. #16 AND [humans]/lim #20. #18 OR #19 #21. (#18 OR #19) AND ([english]/lim) DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 1,010 UPDATED DATE OF SEARCH: February 24, 2021 NUMBER OF ADDITIONAL RESULTS: 60

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Cochrane Library	#1. [mh thyroidectomy] #2. (Thyroidectomy or Thyroidectomies or strumectomy):ti,ab #3. [mh thyroid] or thyroid:ti,ab #4. (resection or surgery or removal or operation or remove):ti,ab,kw #5. #3 and #4 #6. #1 or #2 or #5 #7. [mh osteoporosis] #8. (Osteoporosis or Osteoporoses or Osteoporotic or "Female Athlete Triad" or "Bone Loss" or "Bone Losses"):ti,ab #9. (Osteopenia or Osteopenias):ti,ab #10. [mh "Bone Density"] #11. ("Bone density" or "Bone densities" or "mineral density" or "mineral densities" or "osseous density" or "osseous densities" or "bone mineral content" or "bone mineral contents" or (BMD and bone)):ti,ab #12. [mh "bone fractures"] #13. (fracture or fractures or "broken bone" or "broken bones"):ti,ab #14. {or #7-#13} #15. #6 and #14 DATE OF SEARCH: April 10, 2020 NUMBER OF RESULTS: 39 UPDATED DATE OF SEARCH: February 24, 2021 NUMBER OF ADDITIONAL RESULTS: 2
Web Of Science	#1. TS=(thyroidectomy OR thyroidectomies OR strumectomy) #2. TS=(thyroid AND (resection OR surgery OR removal OR operation OR remove)) #3. #1 OR #2 #4. TS=(osteoporosis OR osteoporoses OR osteoporotic OR "female athlete triad" or "bone loss" OR "bone losses") #5. TS=(osteopenia OR osteopenias) #6. TS=("bone density" OR "bone densities" OR "mineral density" OR "mineral densities" OR "osseous density" OR "osseous densities" OR "bone mineral content" OR "bone mineral contents" OR (BMD AND bone))
	#7. TS=(fracture OR fractures OR "broken bone" OR "broken bones")

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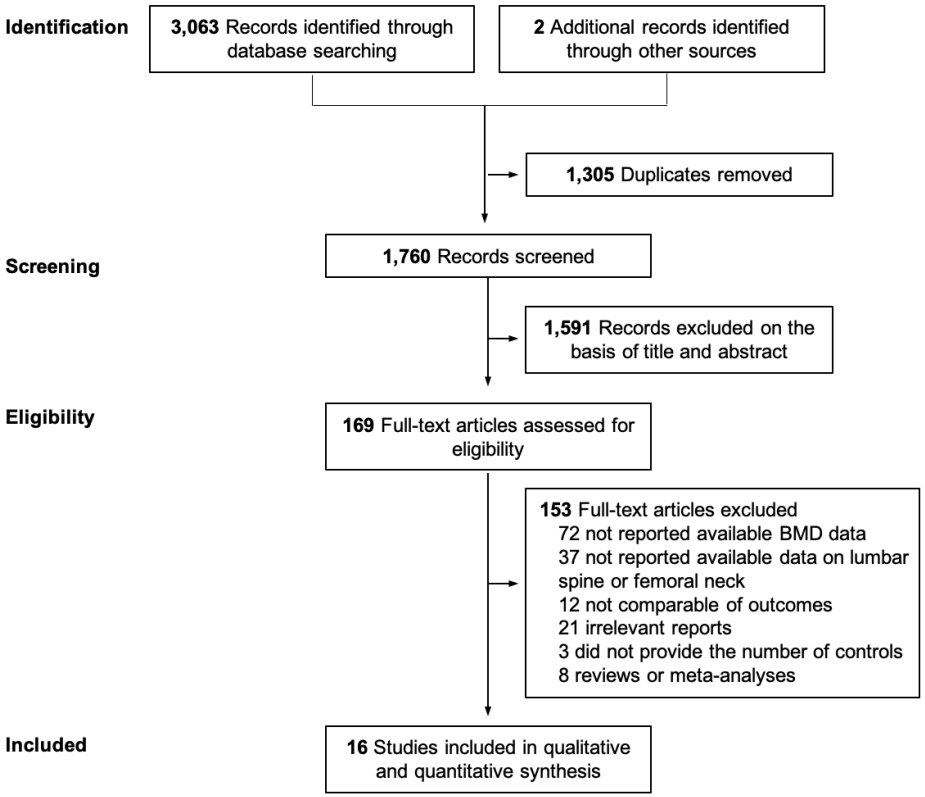
	#8. #4 OR #5 OR #6 OR #7 #9. #3 AND #8 #10. (#3 AND #8) AND LANGUAGE: (English) DATE OF SEARCH: February 24, 2021 NUMBER OF RESULTS: 370
Scopus	#1. TITLE-ABS-KEY(thyroidectomy OR thyroidectomies OR strumectomy) #2. TITLE-ABS-KEY(thyroid AND (resection OR surgery OR removal OR operation OR remove)) #3. #1 OR #2 #4. TITLE-ABS-KEY(osteoporosis OR osteoporoses OR osteoporotic OR "female athlete triad" or "bone loss" OR "bone losses") #5. TITLE-ABS-KEY(osteopenia OR osteopenias) #6. TITLE-ABS-KEY("bone density" OR "bone densities" OR "mineral density" OR "mineral densities" OR "osseous density" OR "osseous densities" OR "bone mineral content" OR "bone mineral contents" OR (BMD AND bone)) #7. TITLE-ABS-KEY(fracture OR fractures OR "broken bone" OR "broken bones") #8. #4 OR #5 OR #6 OR #7 #9. #3 AND #8 #10. (#3 AND #8) AND LANGUAGE(English) DATE OF SEARCH: February 24, 2021 NUMBER OF RESULTS: 1,141
Related article search	Related articles searched according to the following references: 1) Quan ML, Pasieka JL, Rorstad O. Bone mineral density in well-differentiated thyroid cancer patients treated with suppressive thyroxine: a systematic overview of the literature. *J Surg Oncol.* 2002;79(1):62-69; discussion 69-70. 2) Heemstra KA, Hamdy NA, Romijn JA, Smit JW. The effects of thyrotropin-suppressive therapy on bone metabolism in patients with well-differentiated thyroid carcinoma. *Thyroid.* 2006;16(6):583-591. 3) Parker WA, Edafe O, Balasubramanian SP. Long-term treatment-related morbidity in differentiated thyroid cancer: a systematic review of the literature. *Pragmat Obs Res.* 2017;8:57-67. 4) Uzzan B, Campos J, Cucherat M, Nony P, Boissel JP, Perret GY. Effects on bone mass of long term treatment with thyroid hormones: a meta-analysis. *J Clin*

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	Endocrinol Metab. 1996;81(12):4278-4289. 5) Yoon B-H, Lee Y, Oh HJ, Kim SH, Lee Y-K. Influence of thyroid-stimulating hormone suppression therapy on bone mineral density in patients with differentiated thyroid cancer: a meta-analysis DATE OF SEARCH FOR RELATED ARTICLES: April 20, 2020 TOTAL: 332 NUMBER OF RESULTS: 2

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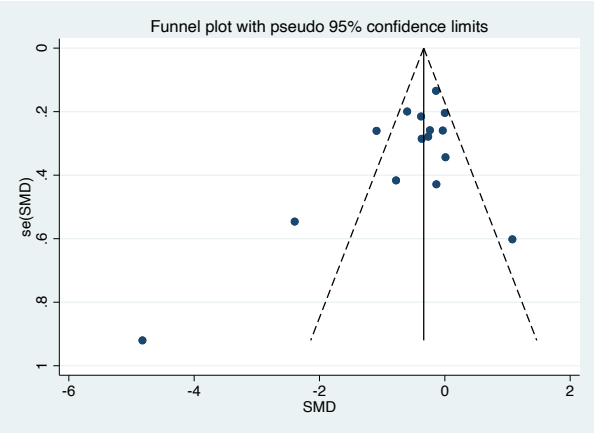
Supplemental Figure 1. Flow diagram of reference selection process



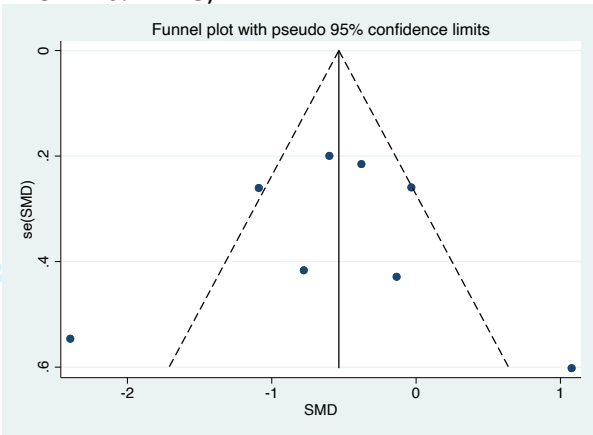
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Supplemental Figure 2a. Funnel plots for evaluation of publication bias (Lumbar spine)

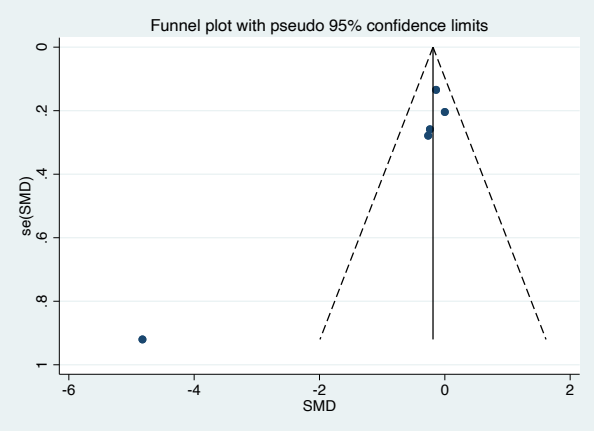
Overall



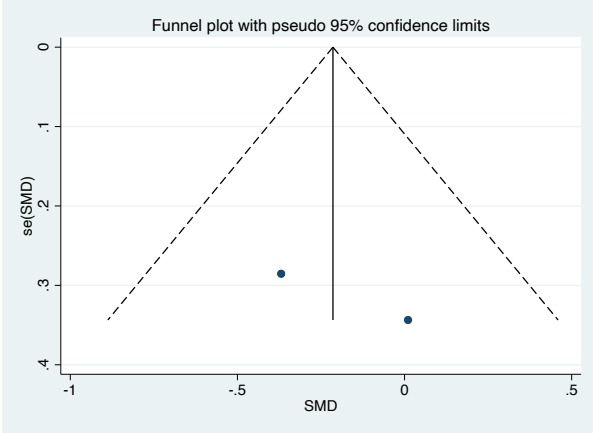
TSH <0.1 mIU/L



TSH 0.1-0.49 mIU/L



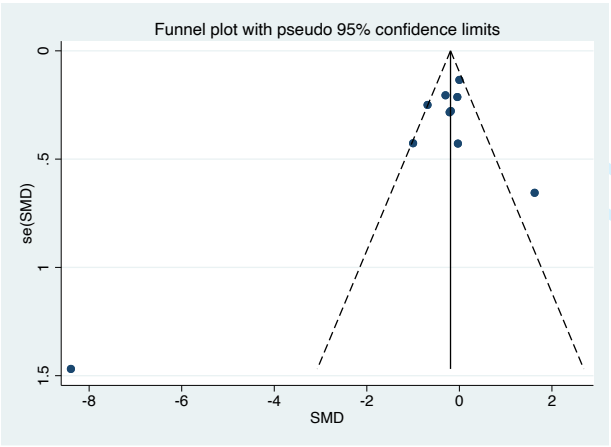
TSH ≥0.50 mIU/L



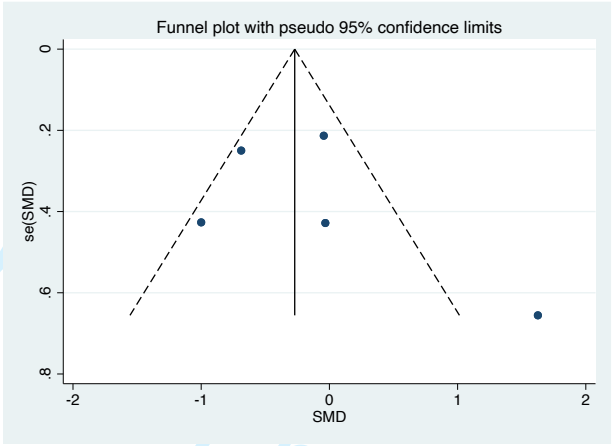
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Supplemental Figure 2b. Funnel plots for evaluation of publication bias (Femoral neck)

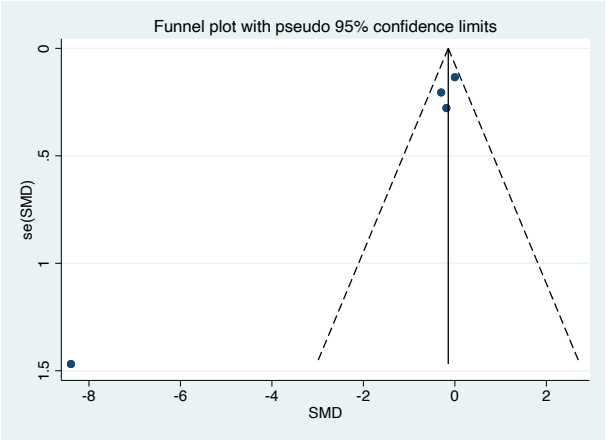
Overall



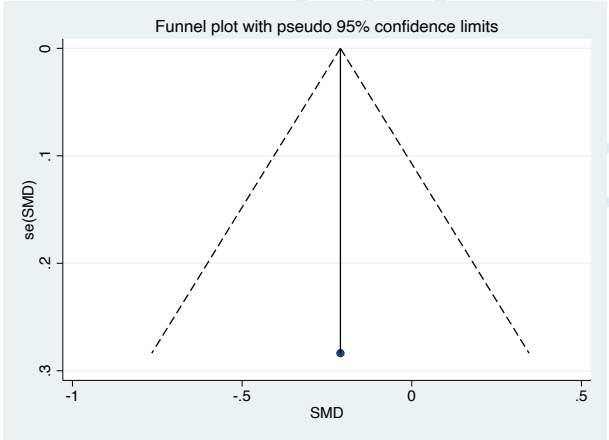
TSH <0.1 mIU/L



TSH 0.1-0.49 mIU/L



TSH ≥0.50 mIU/L



Supplemental Table 2. Additional information on the included studies

First author (year)	Postoperative levothyroxine dose, mean	TSH level at follow-up period, mean	Lists of exclusion criteria for medications*	Lists of exclusion criteria for medical conditions†
Recker (1989)	Range, 0.15-0.20 mg/day	Patients, 1.8 (SD, 0.9) mIU/L; Controls, 3.2 (SD, 1.6) mIU/L	Estrogens, corticosteroids, calcium	Metastatic thyroid cancer, diabetes, post-oophorectomy, post-hysterectomy
Diamond (1991)	Cumulative dose of levothyroxine, mg: 337 (SE, 72)	‡ Patients, 0.3 (SE, 0.1) mIU/L; Controls, 1.2 (SE, 0.2) mIU/L	Estrogens, calcium, vitamin D	(Stated without the list of medical conditions)
Franklyn (1992)	175 (range, 100-200) µg/day	Patients, 0.26 (SD, 0.54) mIU/L; Controls, 1.94 (SD, 1.32) mIU/L	Estrogens, calcium, vitamin D, tamoxifen, thiazide diuretics	Recurrent or metastatic thyroid cancer, osteoporotic fracture, rheumatoid arthritis, diabetes, hypoparathyroidism, alcohol abuse, chronic amenorrhea, late menarche, early menopause, post-oophorectomy
Abugassa (1993)	0.23 (SE, 0.02) mg/day	<0.1 mIU/L	Estrogens	Osteoporotic fracture, endocrinological (apart from hyperparathyroidism), hepatic, intestinal, or renal abnormalities
Kung (1993)	179 (SD, 60) µg/day	Patients, <0.05 mIU/L; Controls, 2.2 (SD, 0.8) mIU/L	Calcium, hormonal contraceptive agents	Recurrent thyroid cancer, late menopause, early menarche, post-oophorectomy, history of other thyroid disorder, family history of osteoporosis
Giannini (1994)	144.2 (SE, 4.15) µg/day	<0.1 mIU/L	(Stated without the list of drugs)	Metastatic thyroid cancer, hypoparathyroidism
Hawkins (1994)	158.3 (SD, 43.7) µg/day	0.3 (SD, 0.4) mIU/L	Estrogens, calcium, vitamin D, thiazide diuretics	(Stated without the list of medical conditions)
Frusciante (1998)	133.6 (SE, 8.4) µg/day	0.04 (SE, 0.016) mIU/L	(Stated without the list of drugs)	(Stated without the list of medical conditions)
Toivonen	215 (SD, 53) µg/day	Patients, <0.05	(Stated without the list of drugs)	(Stated without the list of medical conditions)

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(1998)		mIU/L; Controls, 2.4 (range, 0.06-4.1) mIU/L		
Goerres (1998)	NR ††	0.019 (SD, 0.056) mIU/L	Calcitonin, calcium, vitamin D compounds	NR
Chen (2004)	Starting dose: young patients, 2.0 µg/kg/day; elderly patients, 1.5 µg/kg/day **	1.76 (SD, 0.41) mIU/L	Hormone replacement therapy, corticosteroids, tamoxifen, anticonvulsants, heparin, lithium	Metastatic or recurrent thyroid cancer, hyperparathyroidism, hypoparathyroidism, gonadal insufficiency, diabetes, Cushing's syndrome, severe liver or kidney disorder, chronic obstructive pulmonary disease, rheumatoid arthritis.
Reverter (2005)	NR	Patients, 0.04 (SD, 0.03) mIU/L; Controls, 1.8 (SD, 1.0) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, bisphosphonates, raloxifene, tamoxifen, thiazide diuretics,	Prolonged immobilization, recent bone fracture, inflammatory osteoarticular disease, diabetes, serum creatinine levels >1.3 mg/dL, increased alkaline phosphatase
Eftekhari (2008)	168.33 (SD, 24.3) µg/day	< 0.3 mIU/L	Estrogens, corticosteroids, calcium, vitamin D, bisphosphonates, tamoxifen, thiazide diuretics, anticonvulsants, heparin,	< 45 years, osteoporotic fracture, diabetes, alcohol abuse, rheumatoid arthritis, chronic amenorrhea (> 3 months), late menarche, early menopause, post oophorectomy
Tournis (2015)	122.9 (SD, 21.8) µg/day	Patients, 0.11 (SD, 0.2) mIU/L; Controls, 2.28 (SD, 1.2) mIU/L	Hormonal replacement therapy, corticosteroids, calcium, vitamin D, vitamin D analogues, bisphosphonates, SERMS, denosumab, teriparatide, strontium ranelate, oral contraceptives, diuretics, lithium	Liver or renal disease, prior hyperthyroidism, hyperparathyroidism, hypoparathyroidism, malabsorption syndrome and rheumatic diseases, previous osteoporotic fracture
Moon (2016)	133.5 (SD, 33.5) µg/day	0.09 (SD, 0.13) mIU/L	Bisphosphonate, oral contraceptives, menopausal hormone therapy, SERMS, diuretics, lithium, corticosteroids	Liver or renal diseases, hyperthyroidism, hyper- or hypoparathyroidism, malabsorption syndrome, rheumatic diseases
Kim (2019)	135 (SD, 3.06) µg/day	Patients, 0.000 (IQR 0.00 to 0.242); Controls, 2.450 (IQR 1.263 to 3.813)	Glucocorticoids, anti-osteoporotic drugs	Renal or hepatic impairment

TSH = thyroid stimulating hormone; NR = not reported; NA = not applicable; SD = standard deviation; SE = standard error; SERMS = selective estrogen receptor modulators; hypoPT = patients with hypoparathyroidism; DTC = differentiated thyroid carcinoma; IQR = interquartile range; NTG = non-toxic goiter

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[*] Sixteen studies presented the list of excluded medications, and 6 studies stated that patients who had taken any medication influencing bone density were excluded from the study.

[†] Sixteen studies presented the list of excluded comorbidities, and 6 studies stated that patients who had any medical condition influencing bone density were excluded from the study.

[‡] Lower limit of detection for TSH in this assay was 0.3 mIU/L.

[§] Lower limit of detection for TSH in this assay was 0.15 mIU/L.

[‖] Among 50 participants, 12 were treated with dihydrotachysterol and another 2 with calcitriol, besides calcium tablets. Two patients had well-controlled type 2 diabetes. Three women had been taking hormonal contraceptives for 1 or 3 years, and another had a received low-dose post-menopausal estrogen substitution.

[¶] Lower limit of detection for TSH in this study was 0.2 mIU/L.

[††] Mean cumulated T4 dose of patients was reported as 8527.3 (6102.6) µg/kg body weight.

[**] The dose was adjusted at an increment/decrement of 12.5 to 25 µg/day according to the TSH value at 8- to 12-week intervals.

[‡‡] Levothyroxine dose was adjusted to keep the TSH level under or equal to 0.1 mIU/L.

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Supplemental Table 3. Quality Assessment of Included Studies in the Meta-analysis Using the Newcastle-Ottawa Quality Assessment Scale

First author (year)	Patient selection				Comparability	Exposure			Sum of score
	Adequate case definition	Representativeness of the cases	Selection of controls	Definition of controls		Assessment of Exposure	Same method of ascertainment	Non-Response rate	
Recker (1989)	*	*	*	*	*	-	-	*	6
Diamond (1991)	*	*	*	*	*	-	-	*	6
Franklyn (1992)	*	*	*	*	*	-	-	*	6
Abugassa (1993)	*	*	*	*	**	-	-	*	7
Kung (1993)	*	*	*	*	*	-	-	*	6
Giannini (1994)	*	*	*	*	*	*	-	*	7
Hawkins (1994)	*	*	*	*	*	-	-	*	6
Frusciante (1998)	*	*	*	*	*	*	-	*	7
Toivonen (1998)	*	*	*	*	*	*	-	*	7
Goerres (1998)	*	*	*	*	*	*	-	*	7
Chen (2004)	*	*	*	*	*	*	-	*	7
Reverter (2005)	*	*	*	*	**	*	*	*	9
Eftekhari (2008)	*	*	*	*	**	*	*	*	9
Tournis (2015)	*	*	*	*	**	*	*	*	9
Moon (2016)	*	*	*	*	**	*	*	*	9
Kim (2019)	*	*	*	*	**	*	*	*	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	#1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	#2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	#4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	#5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	#6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	#7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	#6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplemental Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Supplemental Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	#7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	#7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	#8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	#8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	#8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	#8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	#8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Supplemental Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	#9-13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	#18-19
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Figure 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Figure 1
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Figure 2
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	#21
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	#24-25
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	#26
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	#26

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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